

TC 292-MCC: Mechanical Characterization and Structural design of Textile Reinforced Concrete

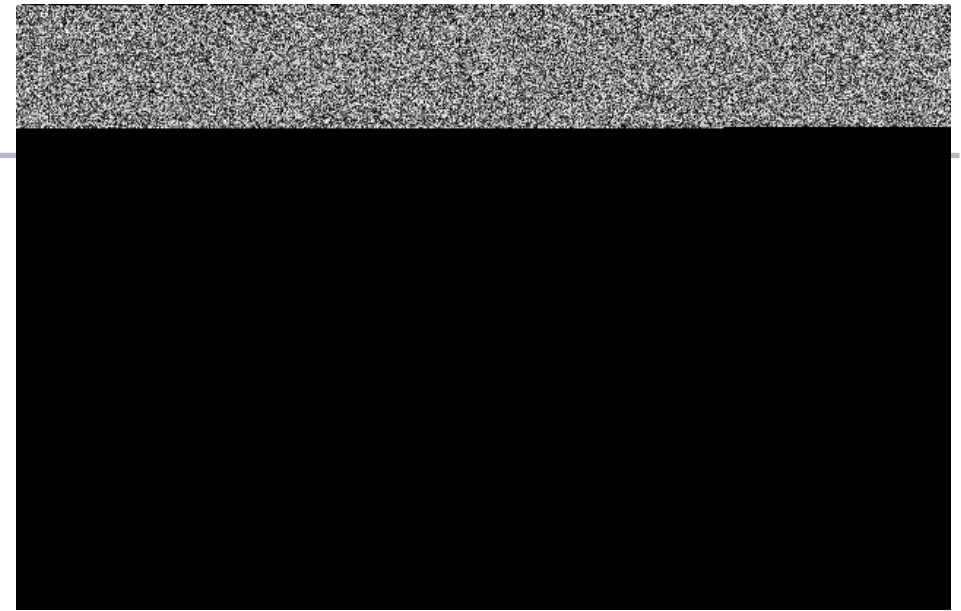
RILEM TC 292 Final Presentation

RILEM 2026 Spring Convention
13-17 April 2026, Ghent, Belgium

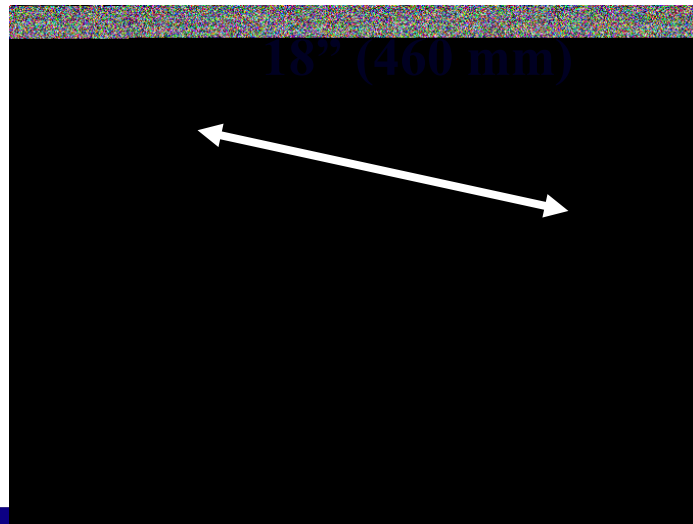
Towards Sustainability and Efficiency by Supplementary Reinforcement

Hypothesis: Strengthening at the microstructure level will have a profound impact at the macro-structural level strength and durability

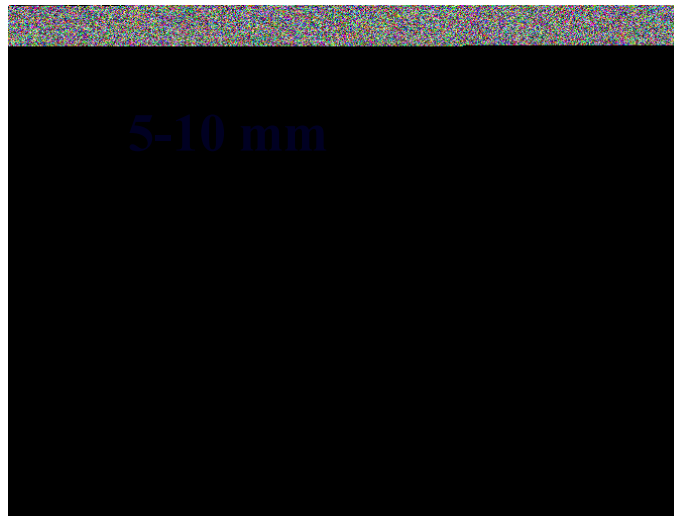
We design a 1 km long bridge using rebars that are 2 cm in diameter. Why couldn't we design a 2 m sized structural element with 20-micron fibers?



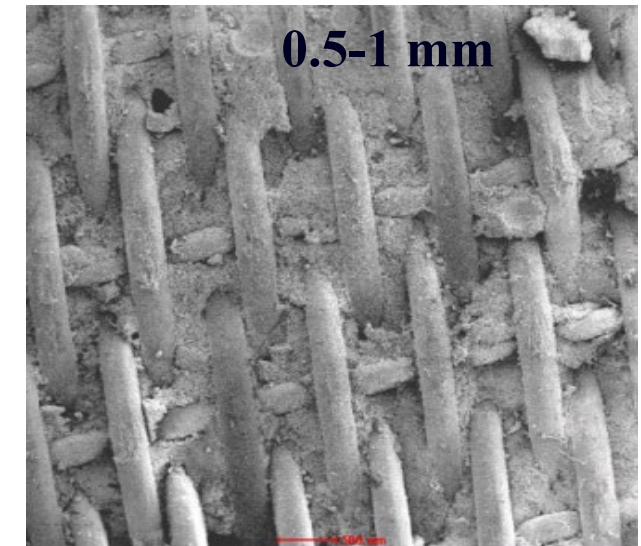
P. L. Nervi 1950's



Ordinary reinforced concrete slab
Macro-structure reinforcement



Fiber reinforced Concrete (FRC}
Meso-structure reinforcement



Textile reinforced Concrete (TRC),
Micro-structure reinforcement

Main topics

1) Textile reinforcement:

- Fibers types
- Coating materials and methods
- Fabric geometry and production

2) Matrix materials and additives:

- Fine grain concrete, smaller than 1mm
- Flowability of fresh mix

3) Manufacturing techniques of TRC:

- Casting/pouring
- Spraying
- Extrusion/pultrusion
- 3D printing

Textile reinforcement

Fiber types [2, 7, 28]

Characteristics	Steel	AR-Glass	Carbon	Basalt	Aramid
<i>Density [g/cm³]</i>	7.85	2.68	1.80	2.6 – 2.8	1.39
<i>Young's modulus [GPa]</i>	210	72	240 – 600	90 – 100	73
<i>Tensile strength [GPa]</i>	0.3 – 0.6	1.0	3.0 – 6.0	4.0	3.4
<i>Corrosion resistance</i>	low	medium	high	low	high
<i>Costs [€/kg]</i>	1 - 3	4 - 6	20 - 30	4 – 6	> 30

Influence of coating [2, 7, 9, 48–50]	Coating types		
	Epoxy coated	SBR coated	uncoated
Tensile strength [MPa] (carbon/AR-glass)	3200 / 1300	2000 / 1100	650-1200 / 450-650
Average value of weight per unit area [g/m ²] (carbon/AR-glass)	455 / 740	369 / 524	300 / 447
Anchorage length ¹	short	medium	medium/long
Dimensional stability ²	high	medium	low
Design flexibility ³	low	medium	high
Feasible geometries ⁴	flat, plane		curved, folded

¹ Fibre length embedded in concrete to ensure load transfer between fibre and concrete.

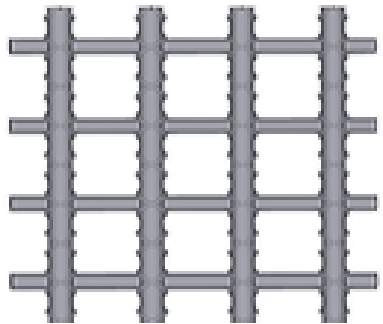
² Measure of the flexural and shear strength of the textile reinforcement

³ Measure of the forming behaviour/drapeability

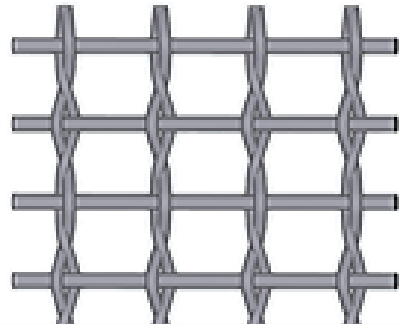
⁴ Possible geometries of the final TRC product

Fabrics geometry

2D grid-like textile reinforcements

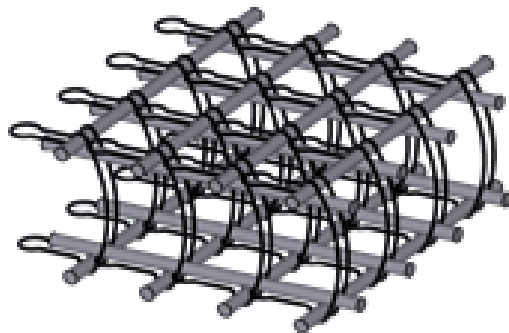


Biaxial warp-knitted

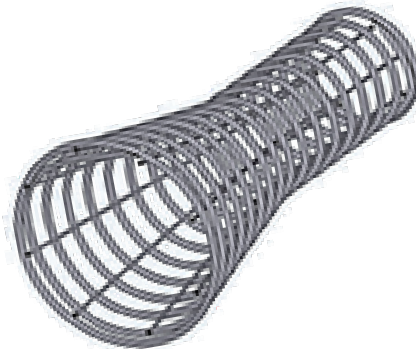


Leno-woven

3D grid-like textile reinforcements

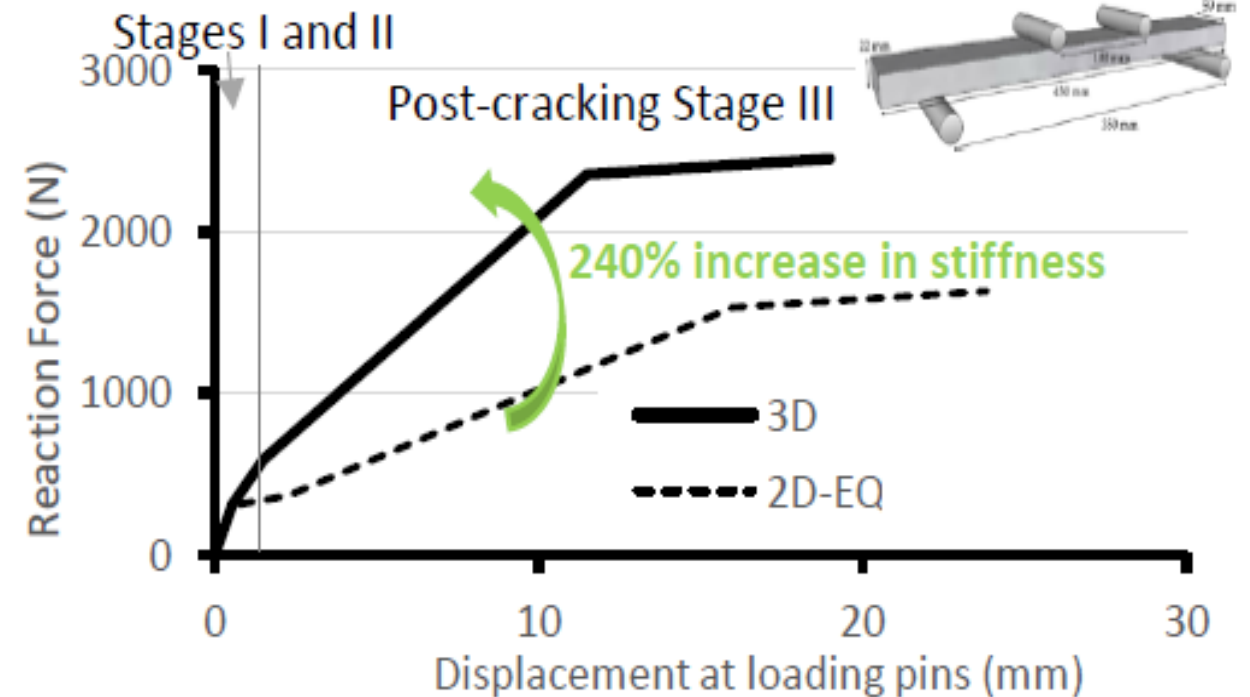


3D-spacer warp-knitted

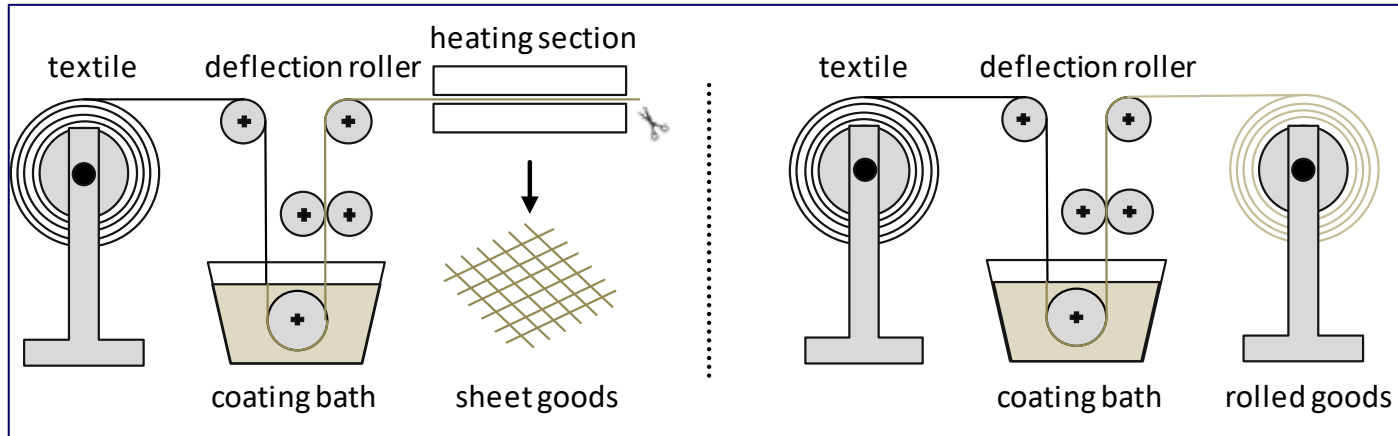


Circular wound

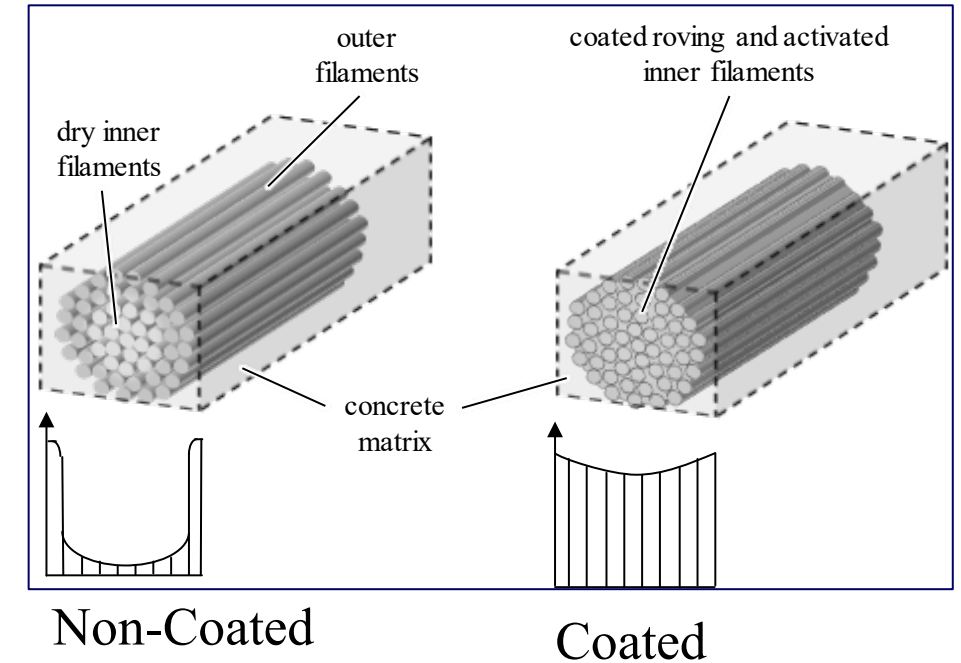
Four point bending of 3D vs. 2D textile fabrics TRC layup [78]



Coating processes [7, 49])



Stress distribution [46])

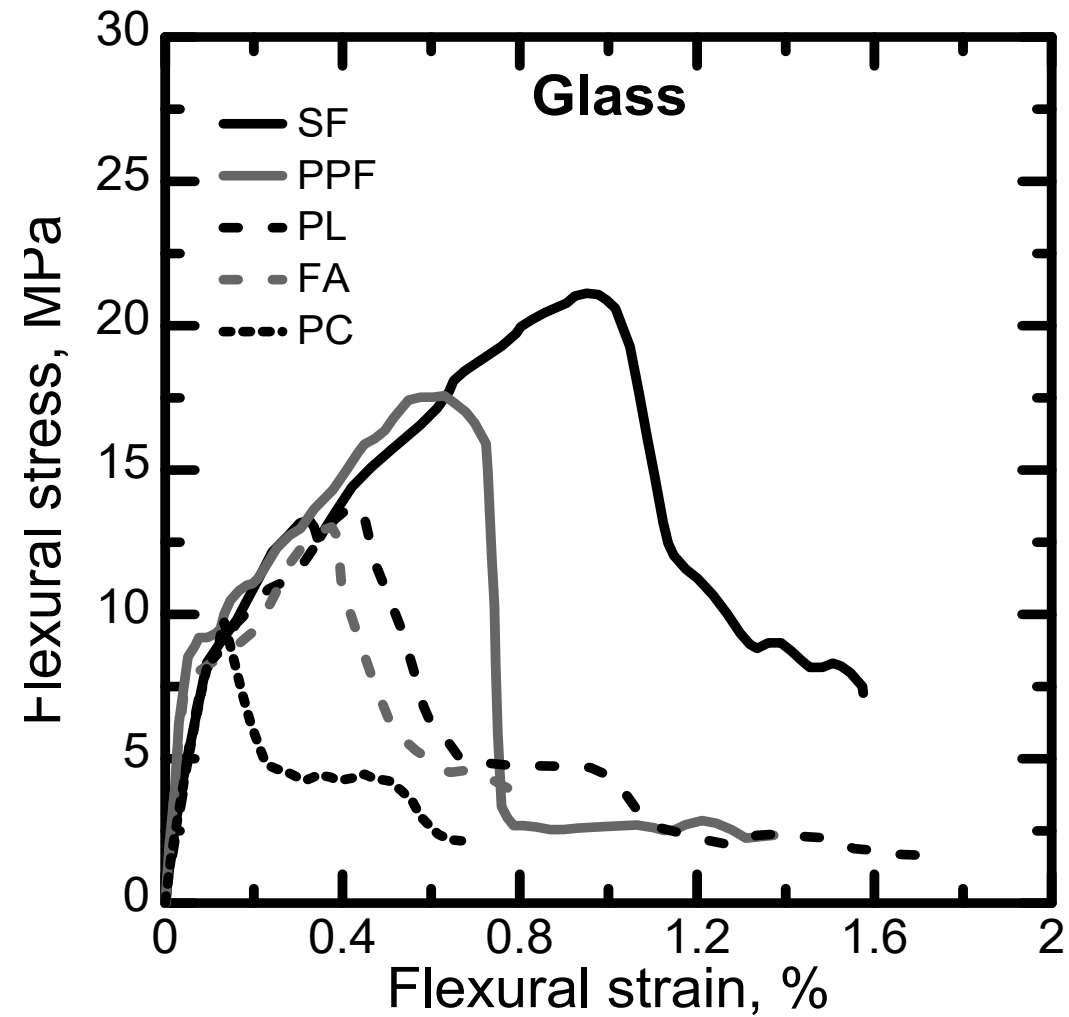


Mass fractions and calculated costs of coating materials [7]

Characteristics	Epoxy resin coating	SBR coating
Mass fraction	183.97 g/m ²	267.62 g/m ²
Cost	183.97 g/m ² · 7.76 €/kg = 1.43 €/m ²	267.62 g/m ² · 2.62 €/kg = 0.70 €/m ²

Influence of additives [29]

- silica fume – SF
- polymer – PL
- short fibres – PPF
- fly ash – FA
- plain matrix– PC



Manufacturing techniques and Applications of TRC

8



a)



b)

[92]



a)

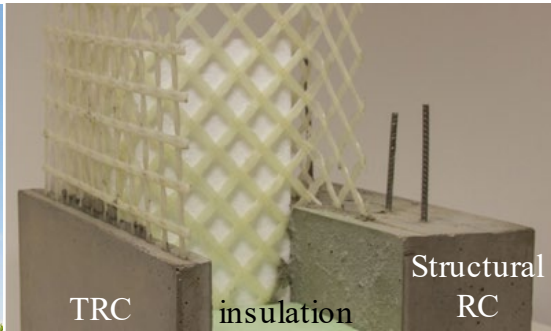


b)

[87]



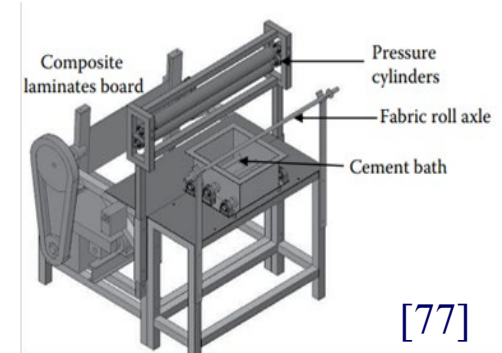
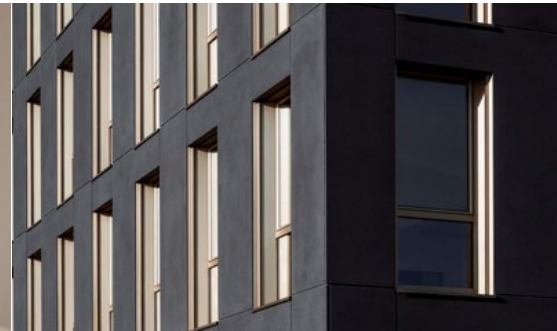
[103]



TRC

insulation

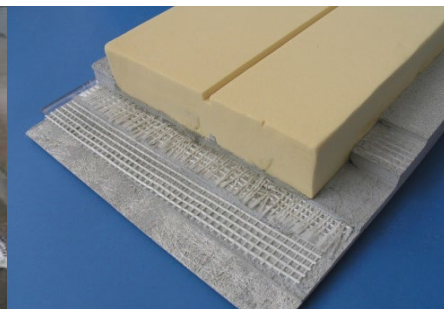
Structural RC



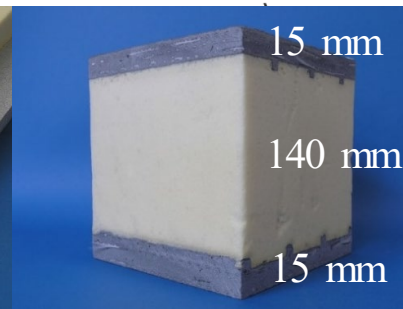
[77]



(a)



(b)



(c)



(d)

[7, 102]

Manufacturing techniques of TRC

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Application areas of the different manufacturing methods [76]

Manufacturing method	Component geometry		Fabric				Reinforcement ratio	Building component
	2D	3D	2D	3D	Uncoated	Coated		
Casting	++	O	++	++	++	++	Reinforcement ratio limited by the density of structure. Spacers required in most cases.	Facades, formwork elements, sandwich elements, structural elements (e.g., diamonds)
Laminating	++	O	++	--	++	o (+)	Reinforcement ratio can be controlled almost arbitrarily.	Facades, slabs
Spraying	++	++	++	--	++	o (+)	Reinforcement ratio can be controlled almost arbitrarily.	Slabs in horizontal and vertical alignment, shells
Extrusion	++	++	Yet unknown					Bar-shaped profiles, honeycomb slabs, formwork elements, hollow core slabs

Main Topics

1 - Tensile and compressive behavior

- Testing Methods
- High Strain Rate
- Use of short fibers and prestressing

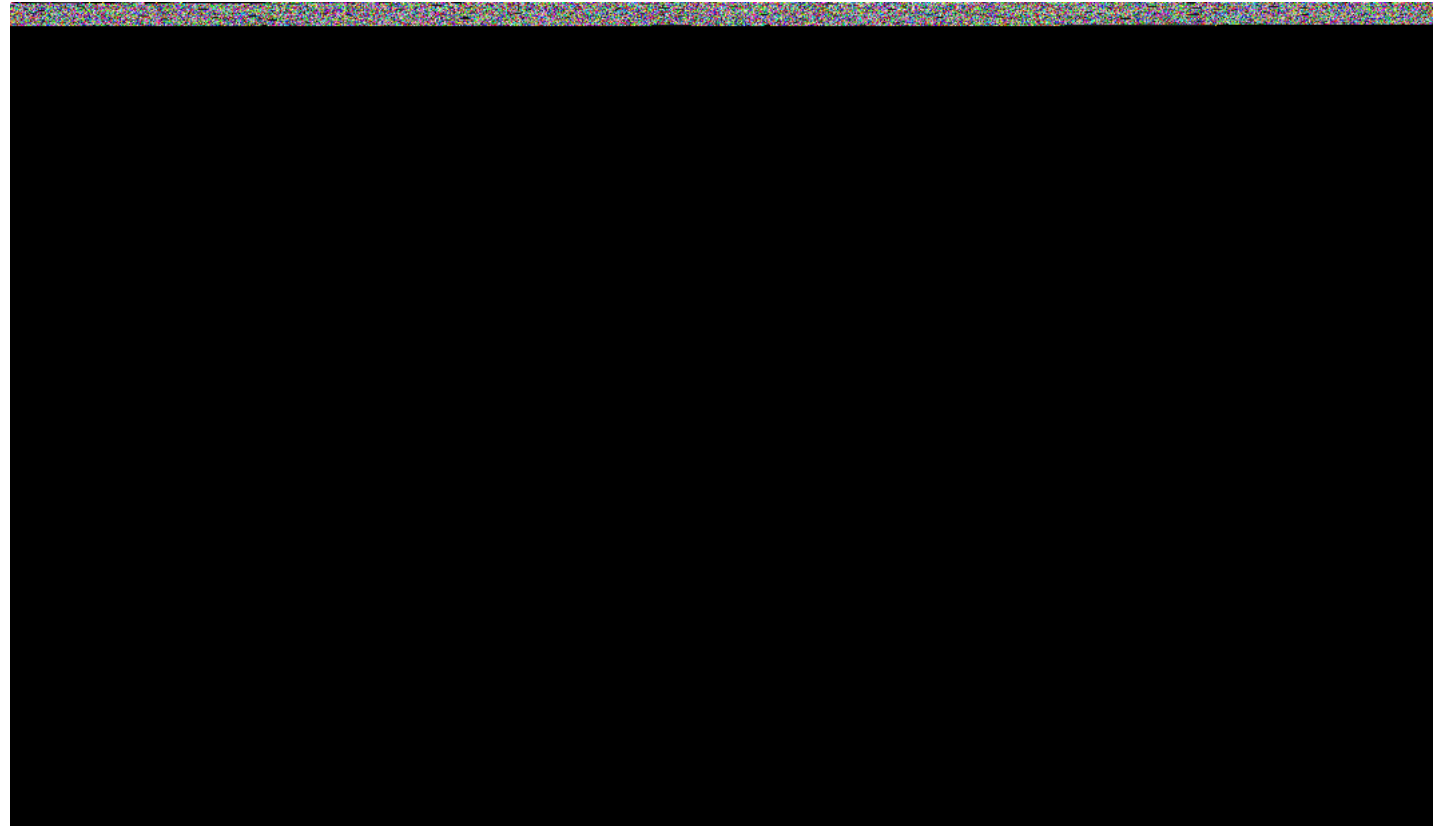
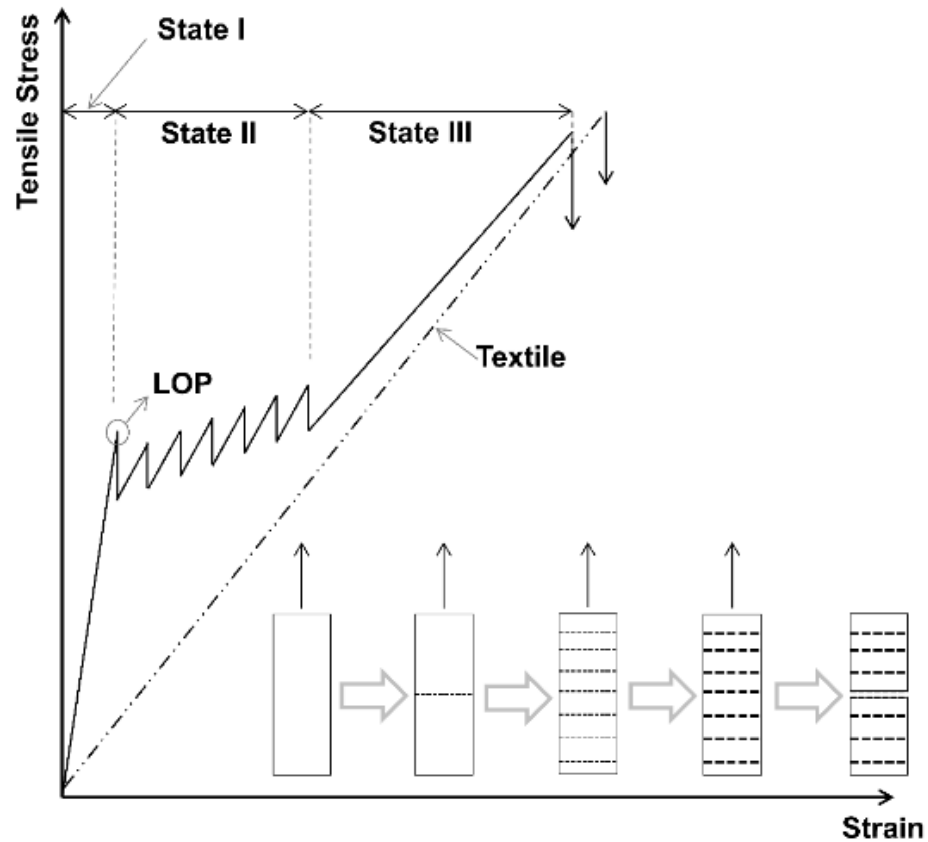
2 – Creep and Fatigue

- Frequency and amplitude effects
- Future work – influence of temperature

3 - Impact

- The use of short fibers
- Instrumented drop weight systems
- Structural testing

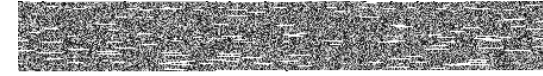
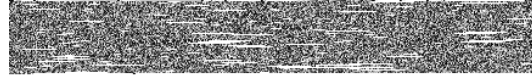
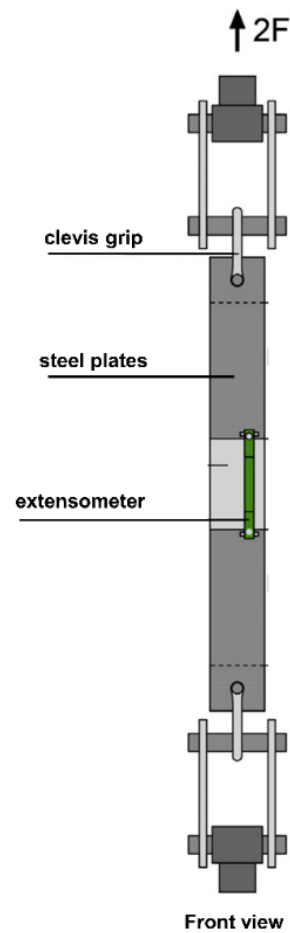
Digital Image Correlation (DIC) to Assess Stress-Crack Width¹¹



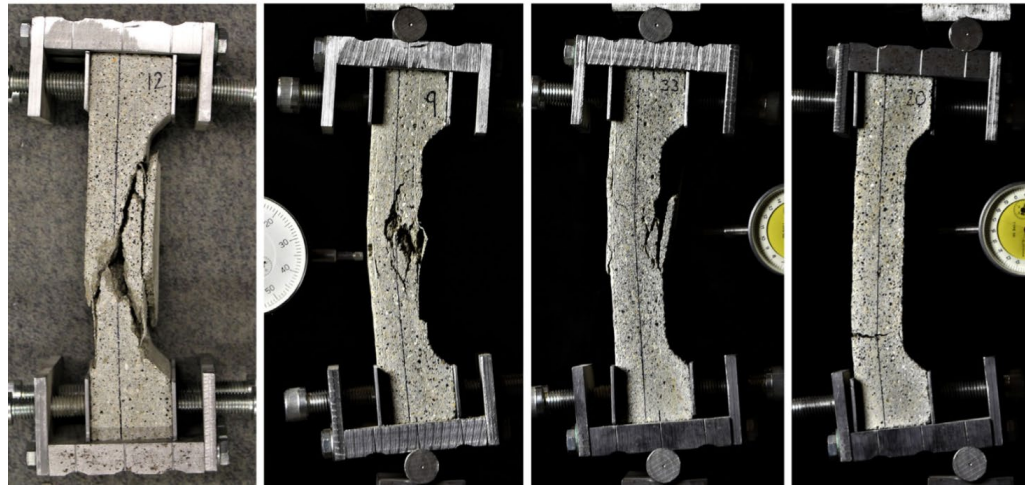
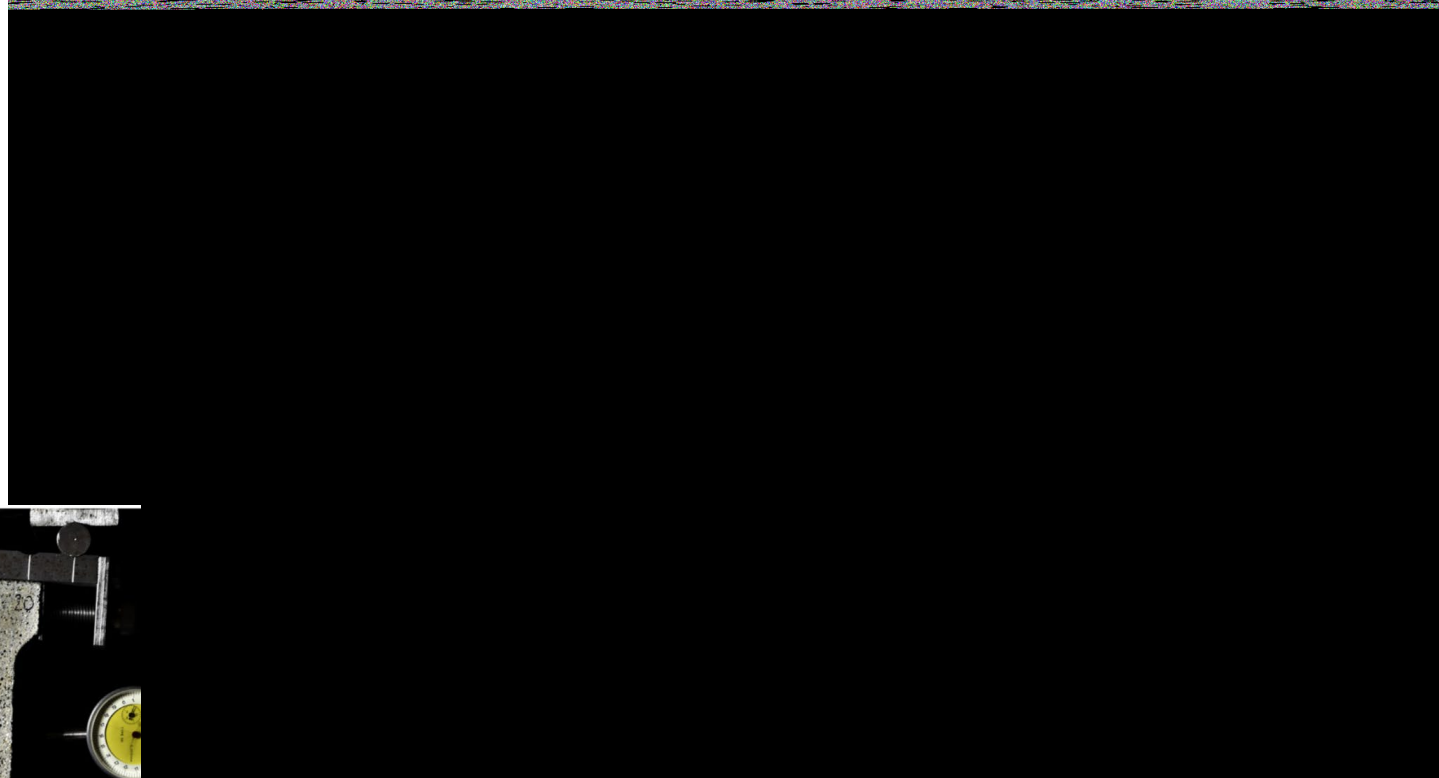
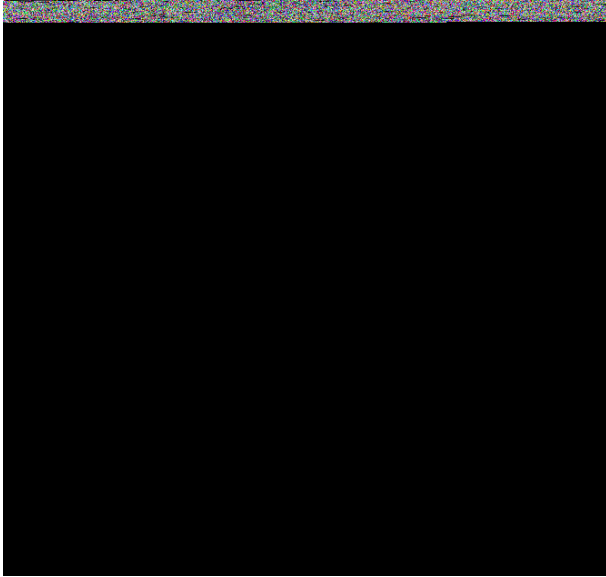
Tensile Test Methods

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Partially clamped (bolts or screws and glue) x Rigid physical boundary conditions (hydraulic Jacks)

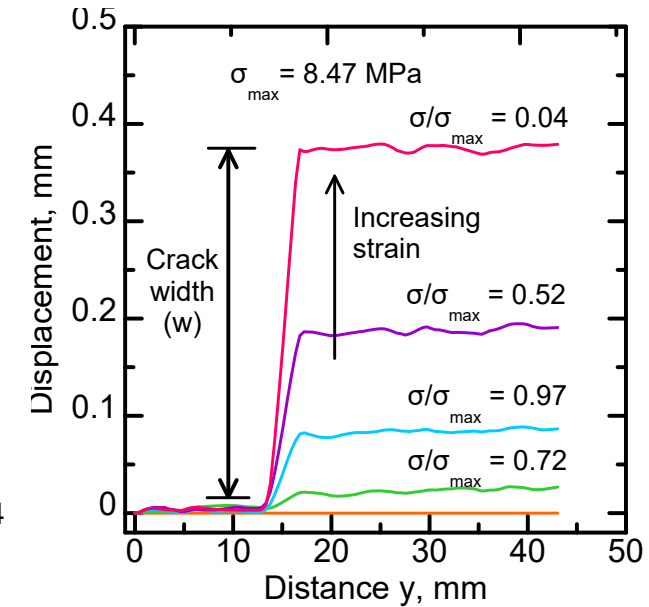
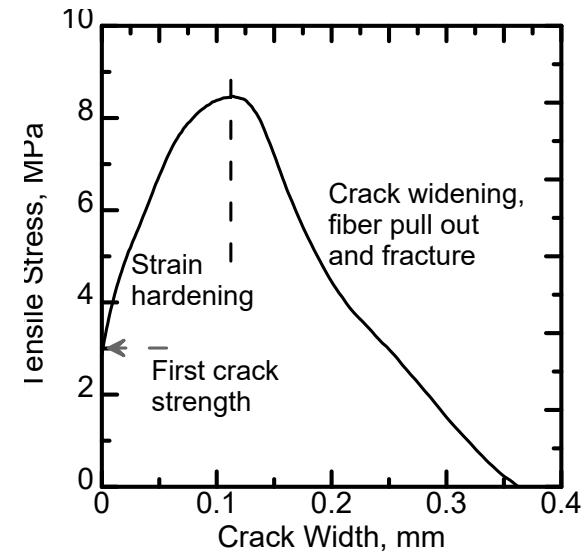
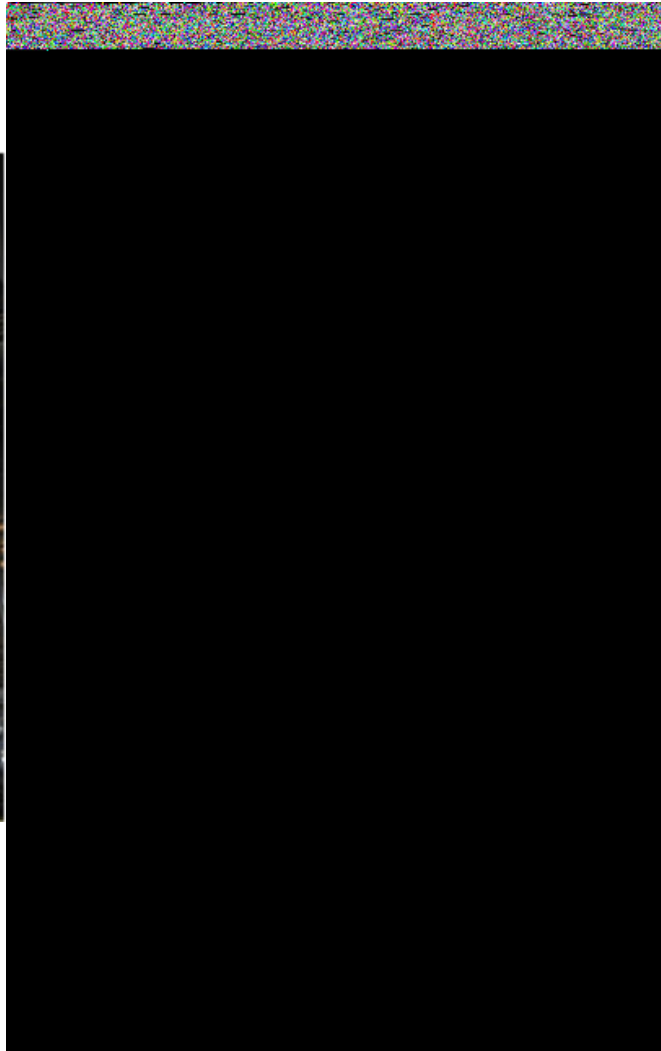


Compressive tests, Structural Design P-M Diagrams for full Domain

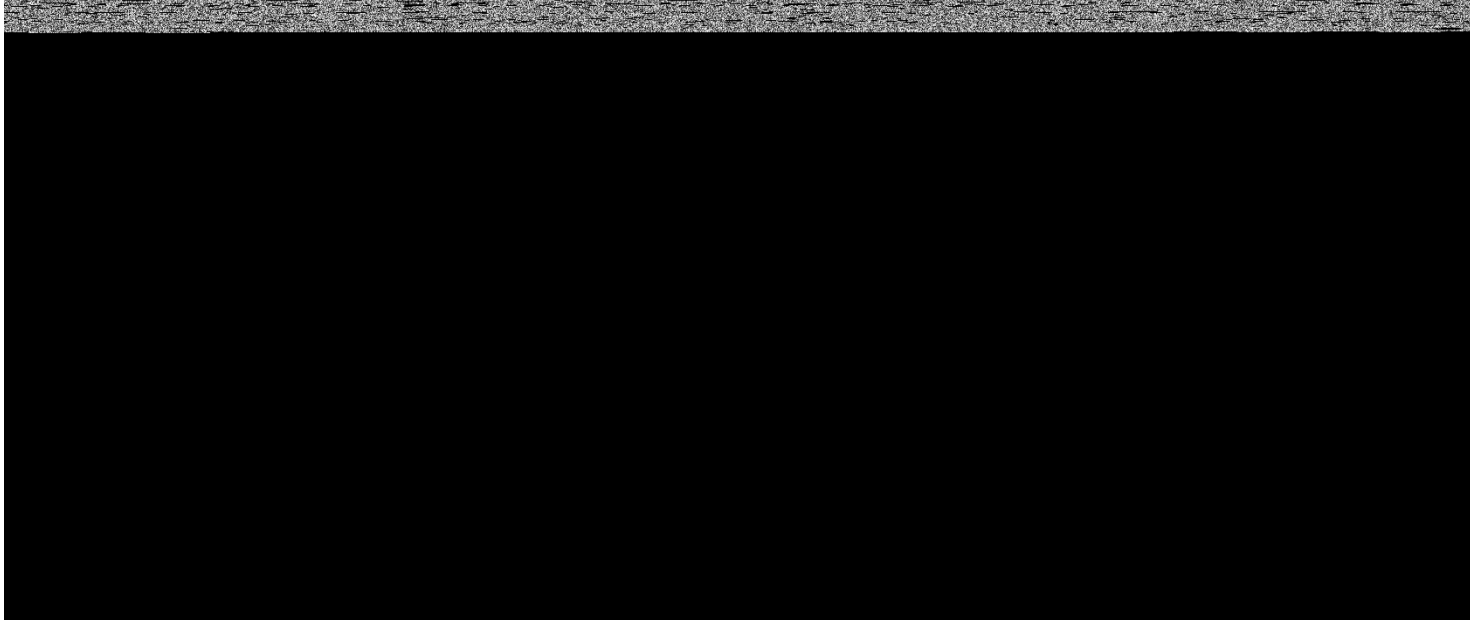


Hawking et al. – Structures 15, 56-65, 2018.

Strain Rate Effects, Interface Characteristics during Failure Mechanisms



Introduction - Significance



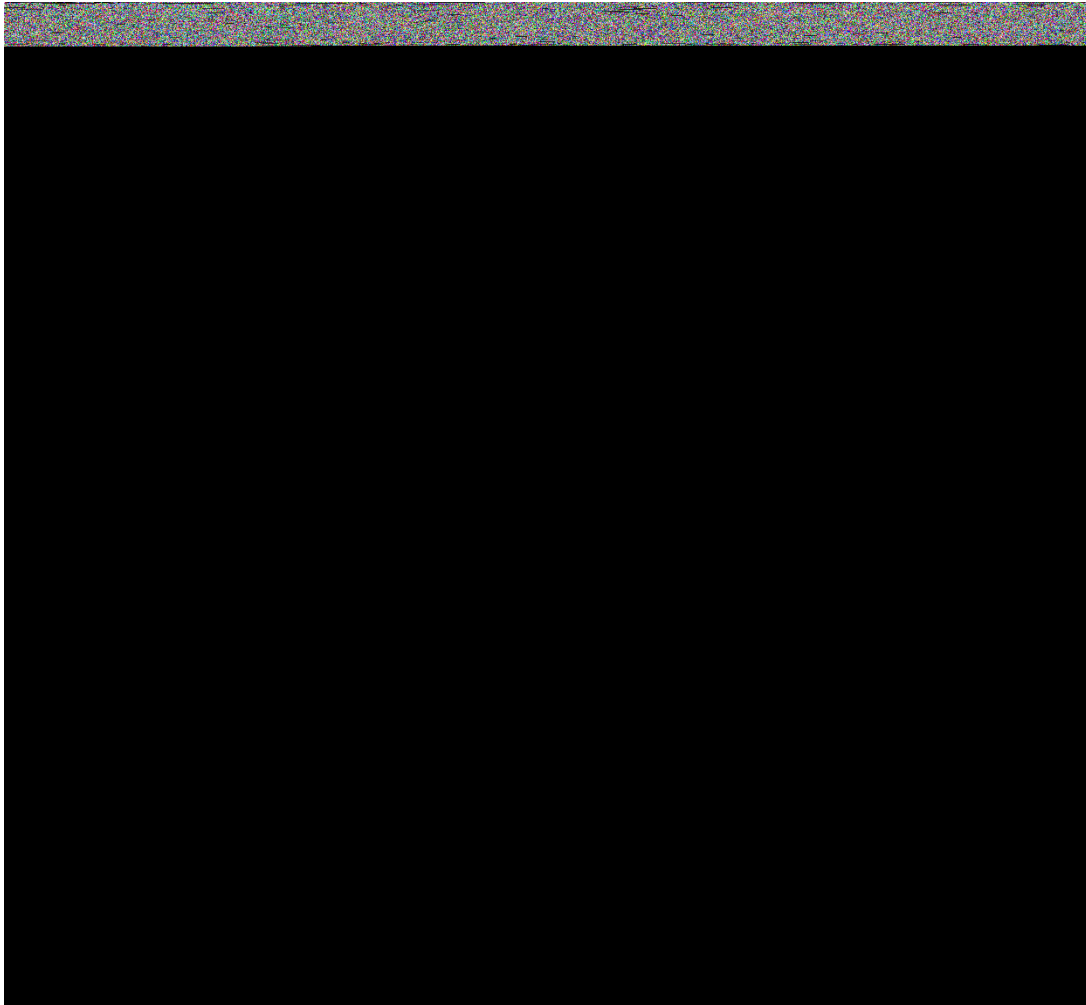
Broad variety of bond performance of TRC has been observed and reported, due to

- Various manufacturing methods
- Lack of standardisation
- TRC still under development



Number of articles on the bond behaviour of TRC published in the years 1970-2022 (data extracted from SCOPUS)

Scope of work on Interface Characterization



- Harmonise terminology for bond-related aspects
- Categorise test configurations and highlight main features of each category.
- Mapping of the complex interconnections of the bond behaviour of TRC

Due to the high significance →

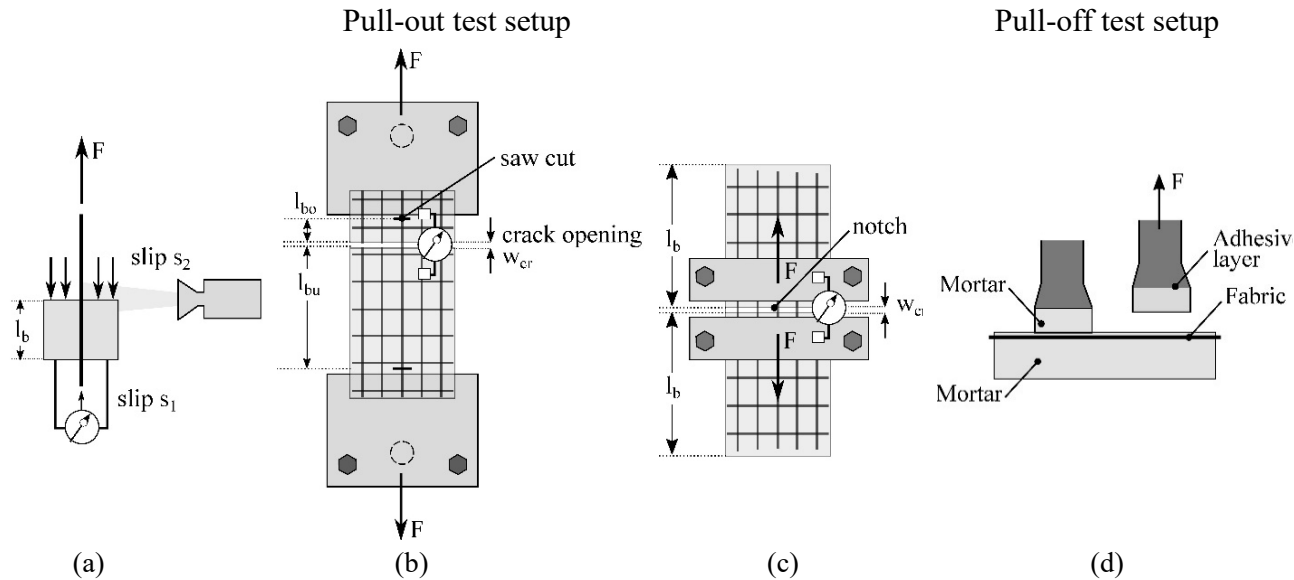
Materials and Structures (2024) 57:97

<https://doi.org/10.1617/s11527-024-02339-5>

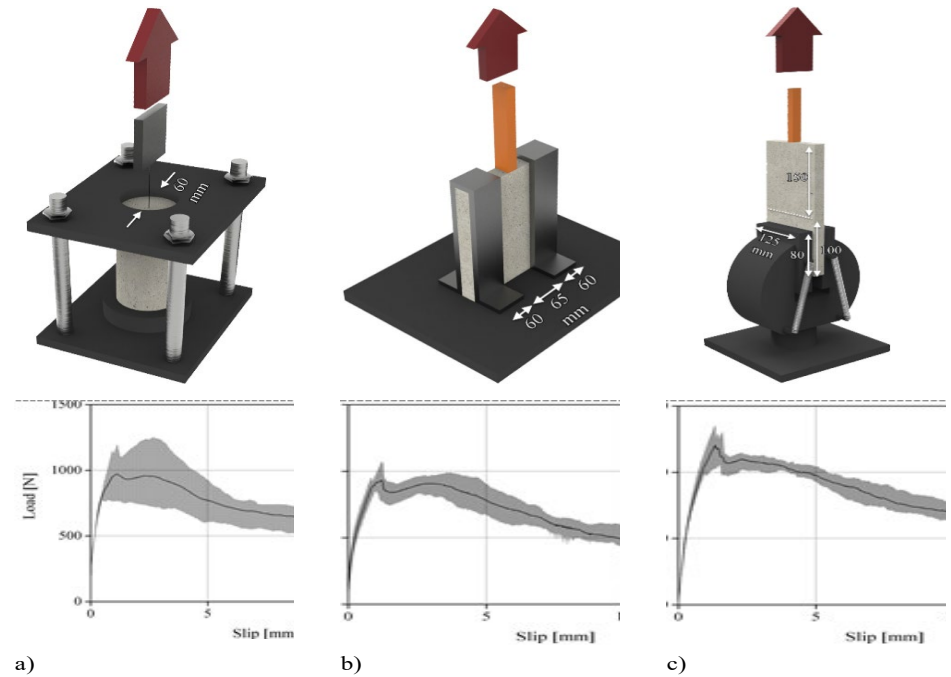
RILEM TC REPORT

Article of RILEM TC 292-MCC: bond behaviour of textile-reinforced concrete—a review

Philipp Preinstorfer^{ID} · Michael El Kadi · Gözdem Dittel · Bahman Ghiassi · Steffen Müller · Rebecca Mansur de Castro Silva · Barzin Mobasher · Flavio de Andrade Silva · Alva Peled



- Four test categories with distinct specifics
- Broad variety within categories that may influence outcome



Dalalbashi A, Ghiassi B, Oliveira D V., Freitas A (2018) Effect of test setup on the fiber-to-mortar pull-out response in TRM composites: <https://doi.org/10.1016/j.compositesb.2018.02.010>



(RILEM TC 292-MCC:
bond behaviour of textile-reinforced
concrete—a review)

- Hierarchical structure: from filament to textile reinforcement
- New developments included, e.g. mineral impregnations or blended cements
- Identification of research gaps

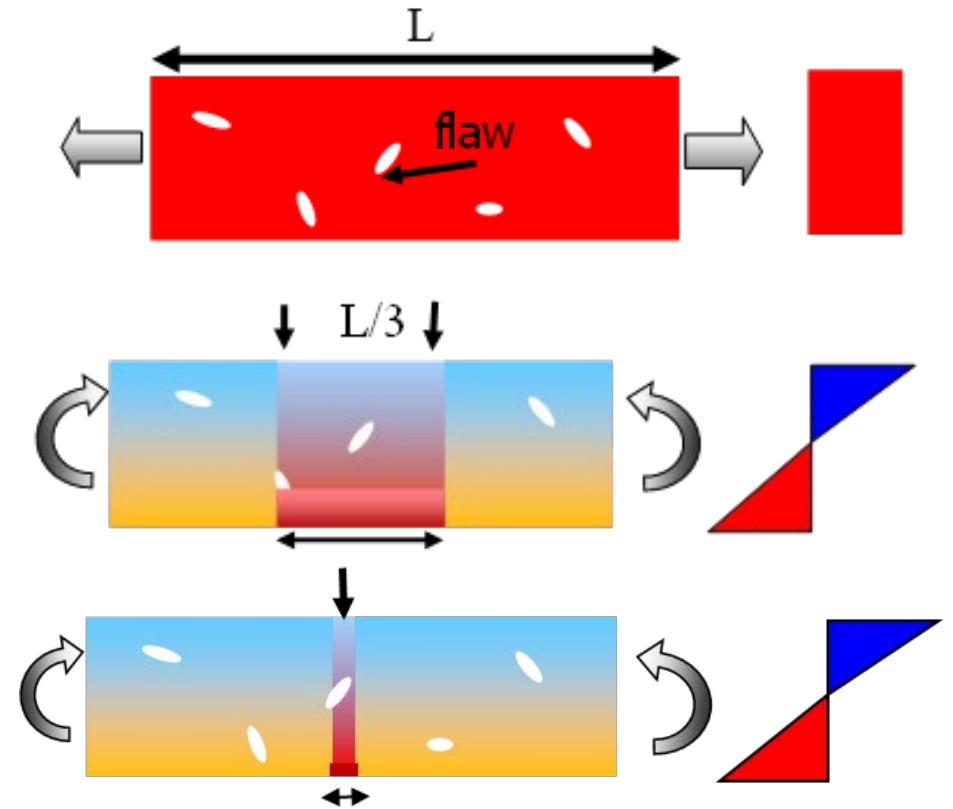
Chapter 5 – Constitutive relationships

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Uniaxial behavior of TRC from a
microstructural size effect POV

Global behavior of TRC
Homogenized constitutive
relationships

Separate models for fabric
and mortar
+
Fabric-mortar interaction

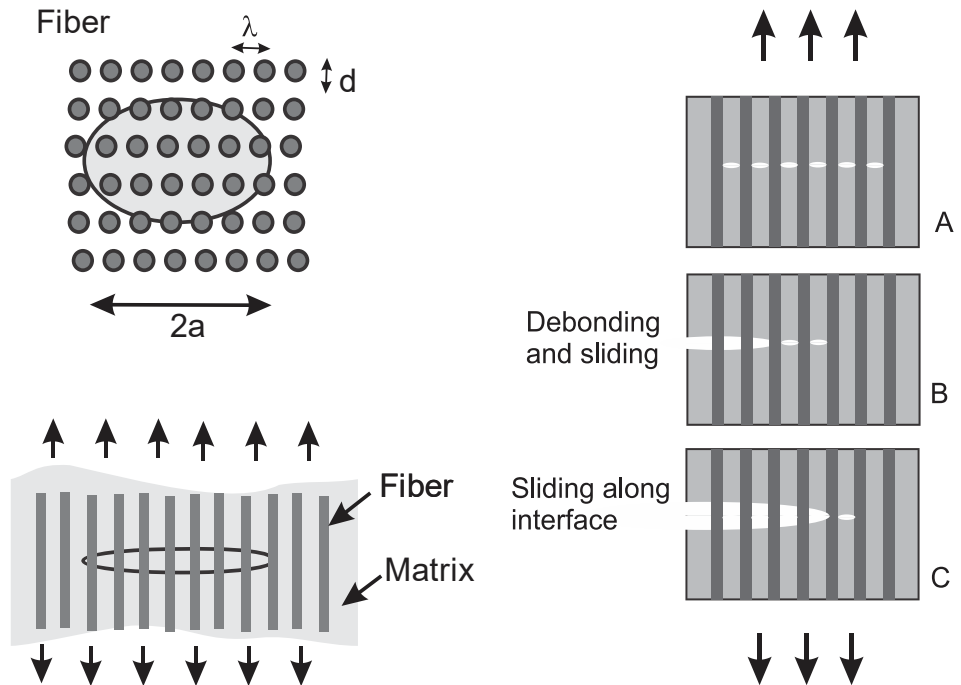


The size of stressed region, and the strain distribution affect the higher strength measures for flexural vs. uniaxial tension.

Use a Weibull approach to address the discrepancy between the tensile and flexural strength of quasi-brittle materials

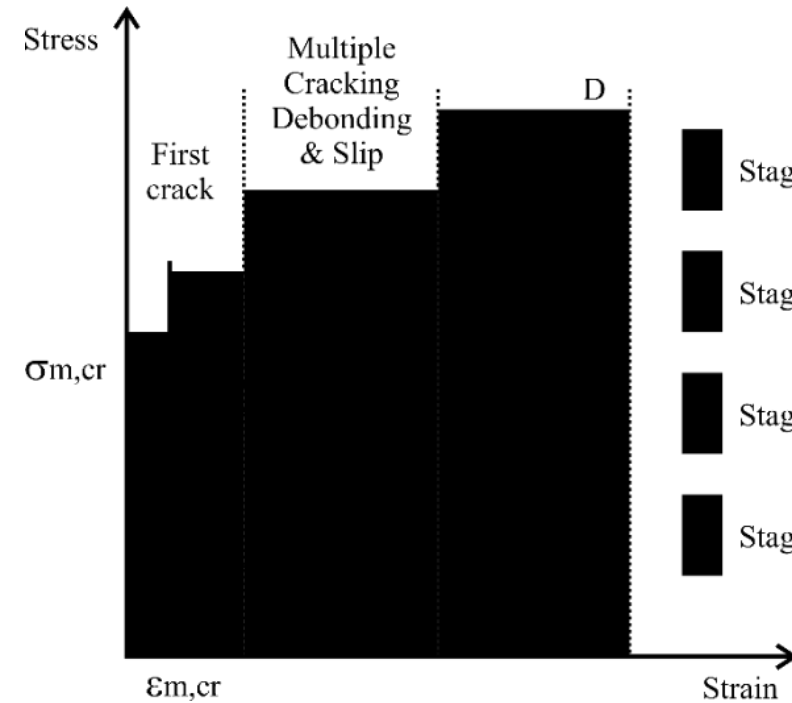
Chapter 5 – Constitutive relationships for Design

20

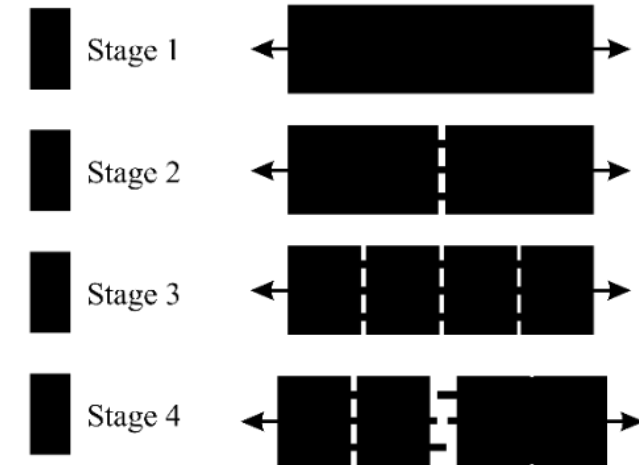


Microcrack initiation and propagate as intersected by fibers.
Fibers bridge matrix cracks through debonding and pullout, hinder crack extension
Efficiency in stress transfer between the fiber and matrix is a source of toughening, formation of multiple cracks, thus improving energy dissipation capacity

(a) Tensile stress-strain evolution



(b) Multiple cracking



Mechanism of crack initiation-propagation, saturation and delamination

Microstructural development of crack propagation

Chapter 5 – Constitutive relationships

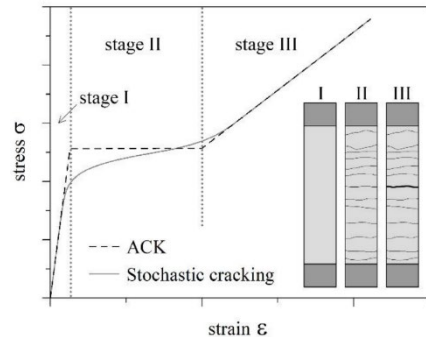
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Homogenized constitutive relationships

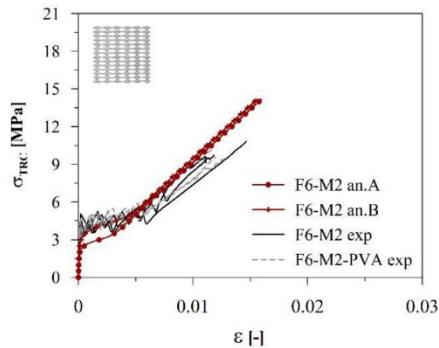
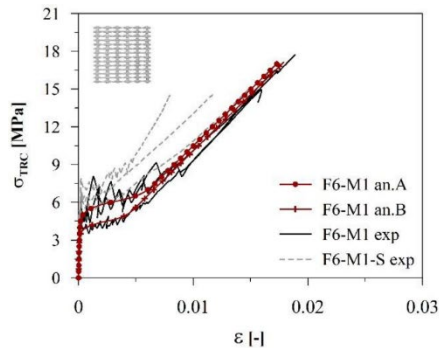
Cuypers and Wastiels 2006

Rampini et al. 2019 – also with short fibres

Stochastic cracking model applied to ACK Theory



Stochastic nature of the tensile strength of the matrix, through the use of a two-parameter Weibull distribution function and same distribution for the crack spacing

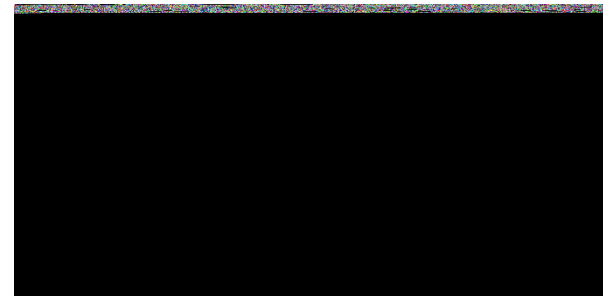


Tri-linear models

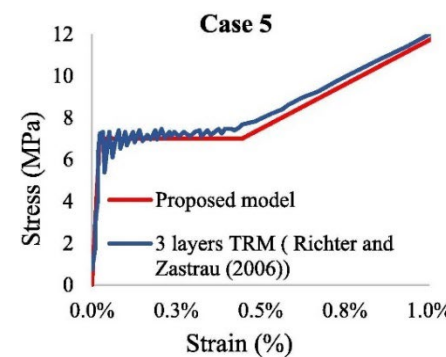
Filippou and Chrysostomou 2020

Larrinaga P, Chastre C, San-José JT, Garmendia L (2013)

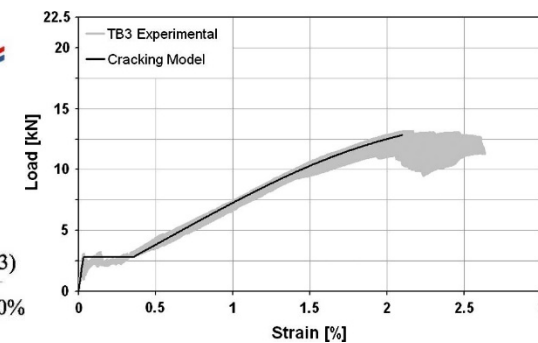
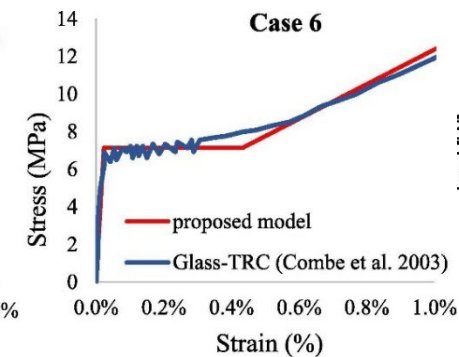
Combination of ACK with crack control expressions used for R/C



Description of Larrinaga model:
non-linear stage III



Validation of Filippou model

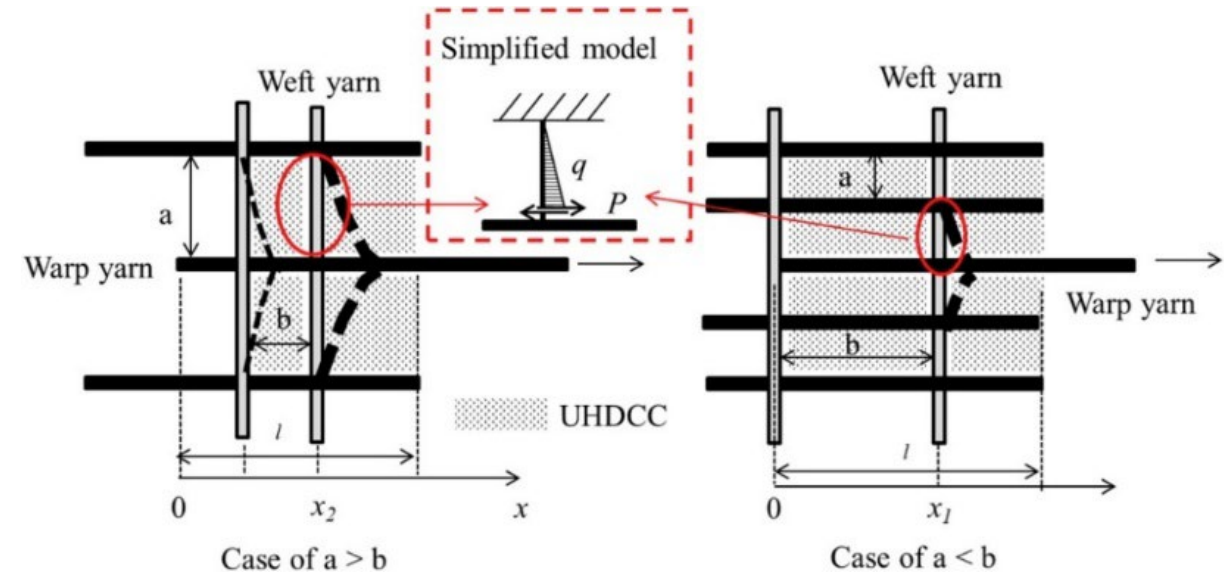


Validation of Larrinaga et al. model

Chapter 5 – Constitutive relationships

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Fabric – mortar interaction



Elastic linear initial phase and subsequent non-linear and dynamic stages

The bond-slip curve in one continuous function: 3 parameters regression of experimental data

Chapter 6 – Special actions

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Constitutive relationships to be used for design of TRC members exposed to special actions

Fire: high temperatures

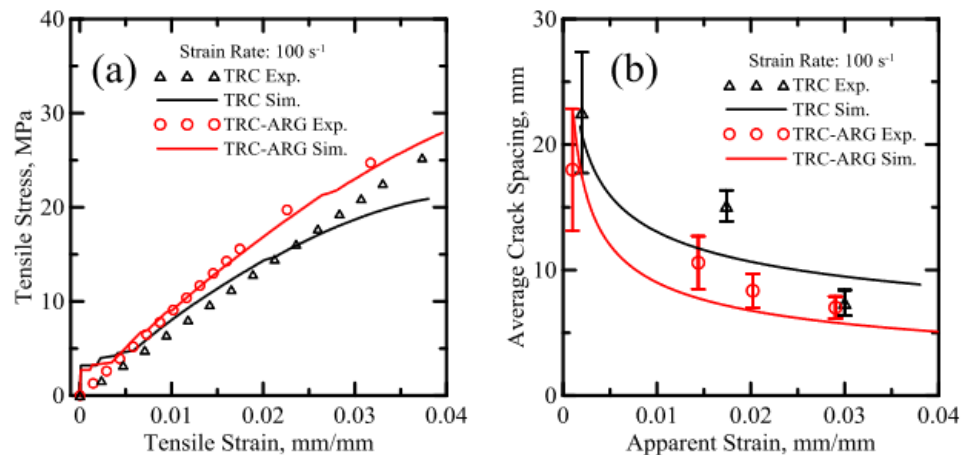
tests on TRC composites at high temperatures and proposed calibration for prediction of the material performance at different temperatures.

Gibson Model-

$$P(T) = R^n \left\{ \frac{P_u + P_r}{2} - \frac{P_u + P_r}{2} \tanh[k_m (T - T_s)] \right\}$$

Bisby Model

$$\frac{P(T)}{P_u} = \frac{(1 - a)}{2} \tanh[-b \cdot (T - c)] + \left(\frac{1 + a}{2} \right)$$

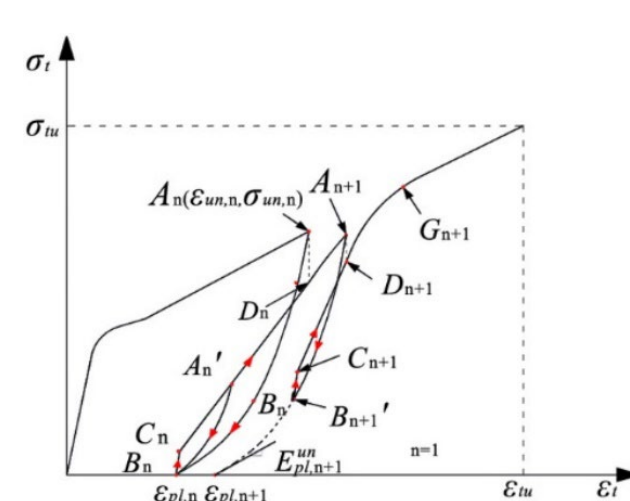


Cyclic loading

Zhu Z-F, Wang W-W, Harries KA, Zheng Y-Z (2018)

From a monotonic approach based on the Richard and Abbot three stage stress strain relationship to model both full or partial unloading and reloading path.

The model accounts for the progressive damage evolution and for the increase of accumulated plastic strain.



Dynamic behavior

Yao Y, Silva FA, Butler M, et al (2015)

Application of finite difference in dynamic conditions

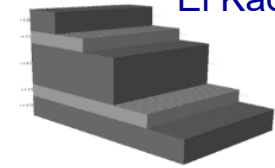
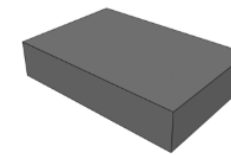
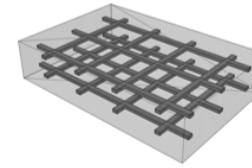
Advances on numerical modelling

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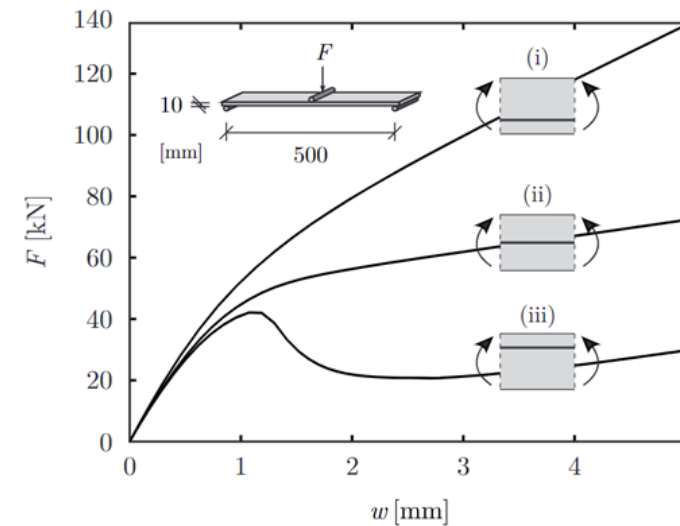
From fully smeared or fully resolved modelling to semi-smeared, layered modelling approaches

→ Computationally-efficient modelling of non-uniformly reinforced sections

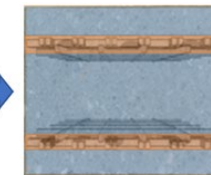
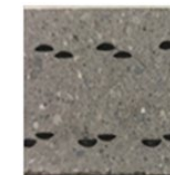
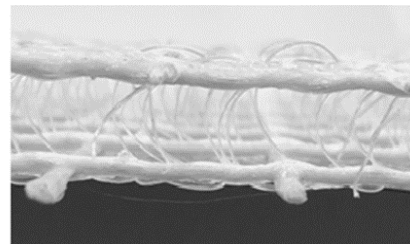
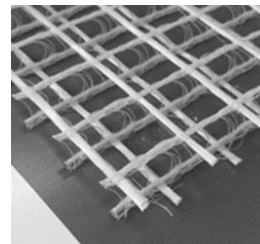
→ Accomodating for material innovations such as 3D spacer textiles



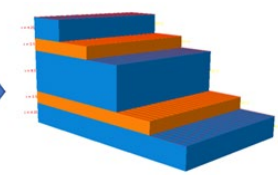
El Kadi (2018)



Chudoba et al. (2019)



El Kadi (2021), El Kadi (2018)



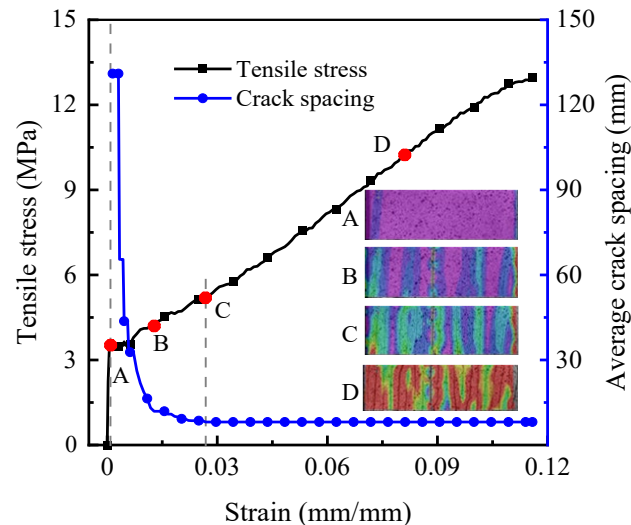
Distributed Cracking Mechanisms in TRC

- Warp knitting pattern (pillar vs. tricot) had no significant impact on tensile properties. Pillar knits recommended due to simpler, less labor-intensive stitching

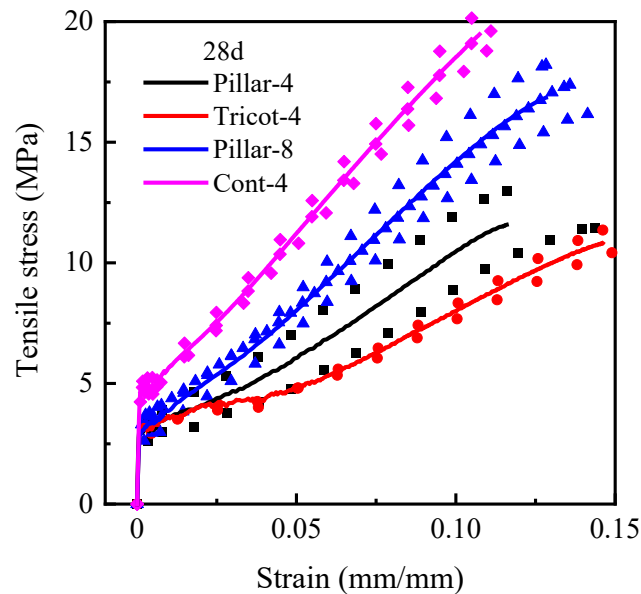
- Doubling textile volume fraction (4% to 8%) increased tensile strength by 44%, post-crack modulus by 78%, and toughness by 47%

- DIC revealed random crack formation rather than sequential spacing reduction along the gage length

- Crack width limit-state of 0.5 mm is satisfied at 75% of ultimate tensile strength (8–12 MPa depending on textile content)



Crack spacing of pultruded pillar TRC specimen (Pillar-4): Stress-Strain vs. Crack location



- DIC tracks each crack via time stamp, strain, and image intensity.
- New cracks form randomly (not sequentially).

1) Constructing force equilibrium to calculate NA

$$\sum F = \varepsilon_{cr} E b \int_{-kh}^{(1-k)h} \sigma(\beta') dy = 0$$

$$0 = \frac{\varepsilon_{cr} E (1-k) b h}{\beta} \left[\int_{k(\beta)}^0 \sigma_c(\beta') d\beta' + \int_0^{\beta} \sigma_t(\beta') d\beta' \right]$$

It is a function of β and material parameters

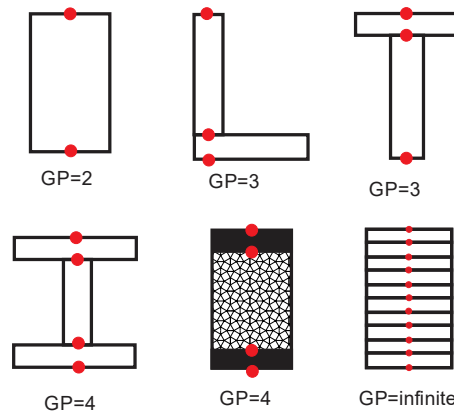
2) Substitute $k(\beta)$ in moment equation

$$M(\beta) = \int_0^h dM(y) = b \int_{(1-k)h}^{-kh} \sigma(\beta') y dy$$

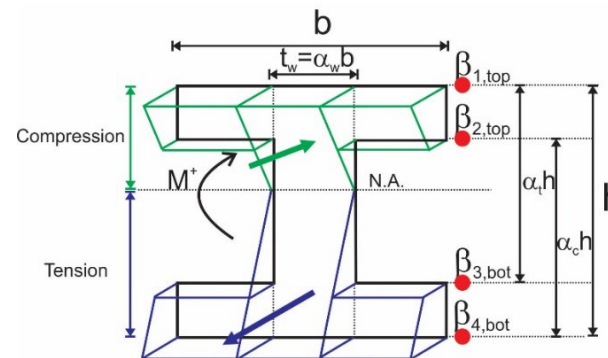
$$= \frac{\varepsilon_{cr} E b h^2 (1-k)^2}{\beta^2} \left\{ \int_0^{\beta} (A_i \beta' + B_i \beta'^2) d\beta' + \int_0^{\lambda} (A_i \beta' + B_i \beta'^2) d\beta' \right\}$$

3) Determine moment at each stage and generate the envelope moment using the boundary condition for each stage

The stress distribution is based on linearized strain at different Stages.



● Geometrical control point, (GP)



$$\varepsilon_{1,top} = \lambda \varepsilon_{cr}$$

$$\varepsilon_{2,top} = \frac{k-1+\alpha_c}{(1-k)} \beta \varepsilon_{cr}$$

$$\varepsilon_{3,bot} = \frac{\alpha_t - k}{(1-k)} \beta \varepsilon_{cr}$$

$$\varepsilon_{4,bot} = \beta \varepsilon_{cr}$$

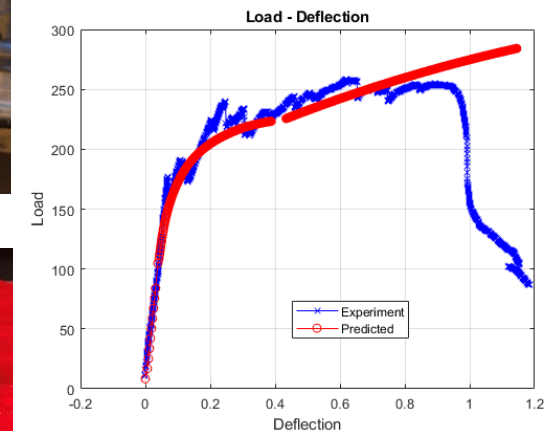
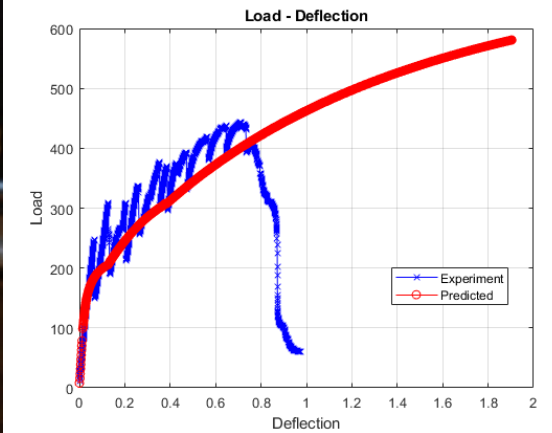
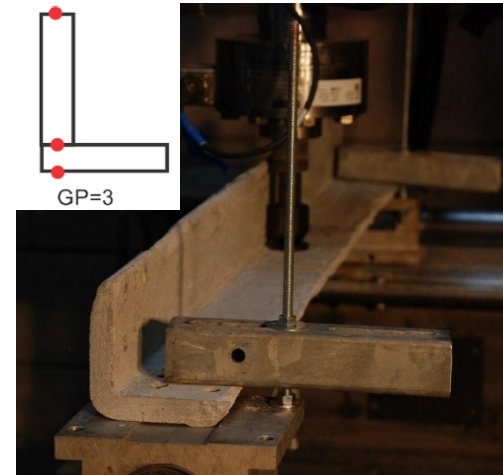
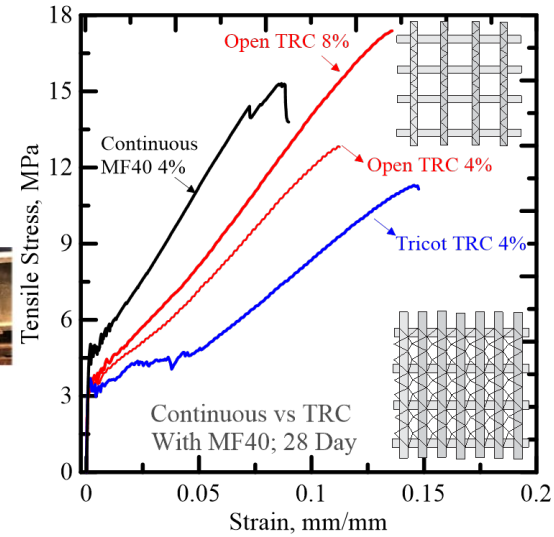
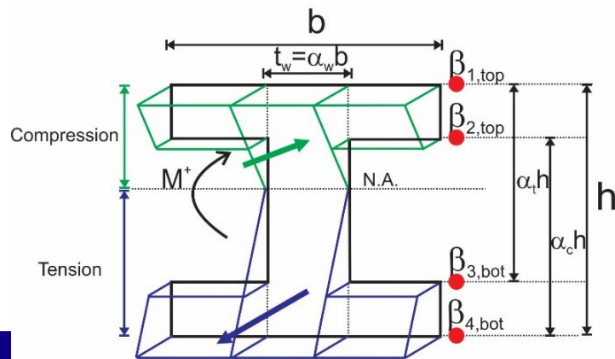
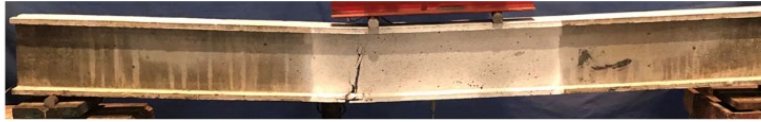
Intersection point connected with logarithmic function of rectangular section

1. Pleesudjai, C., & Mobasher, B. (2023). Analytical moment-curvature solutions for generalized textile-reinforced concrete sections. *Engineering Structures*, 276(May 2022), 115317. <https://doi.org/10.1016/j.engstruct.2022.115317>

2. Mobasher, B., Pleesudjai, C., & Yao, Y. (2021, June). Hierarchical Modeling of Textile Reinforced Concrete Structural Systems. *The 13th International Symposium on Ferrocement and Thin Fiber Reinforced Inorganic Matrices*, Lyon, France

Flexural Modeling for Various Shapes

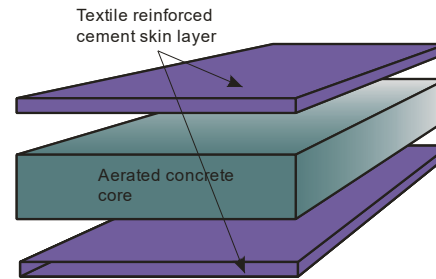
- I-section under Four-point bending test
- 80(b) × 180(d) × 1800(L) mm, t_w and $t_f = 12.5$ mm
- Textile fabrics: Carbon Textile Reinforcement



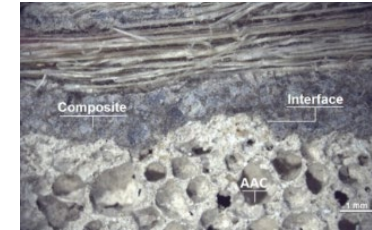
Flexural Modeling of Sandwich Composites

28

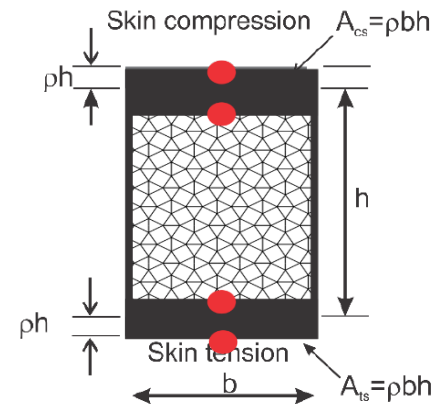
- “TRC-AAC”, brittle core, a shear dominated response, and deflection softening
- “TRC-FRAC”, Ductile core prevented premature shear failure, strain hardening in zone III
- Static test results of: Size A (50x50mm), Size B (50x100)mm, AAC vs FRAC cores
- Damage mechanisms in sandwich composites cannot be explained by a single mode of failure, but the interaction of different modes.



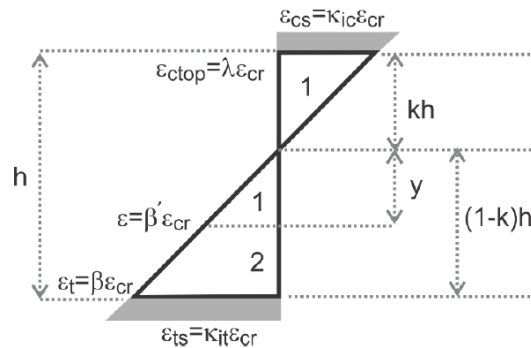
Core material



The plastic length, L_p is the function of considering the number of cracks



Section geometry



Strain distribution

1. Pleesudjai, C., Li, A., Dey, V., & Mobasher, B. (2021). Low-Velocity Impact Experiments and Modeling of TRC Skin Aerated Concrete Core Sandwich Composites. *Materials*, 14(2), 390. <https://doi.org/10.3390/ma14020390>

2. Pleesudjai, C., & Mobasher, B. (2023). Parametric Modeling of Flexural Response of Sandwich Composites. In *Strain Hardening Cementitious Composites. SHCC 2022* (pp. 199–208). Springer International Publishing.

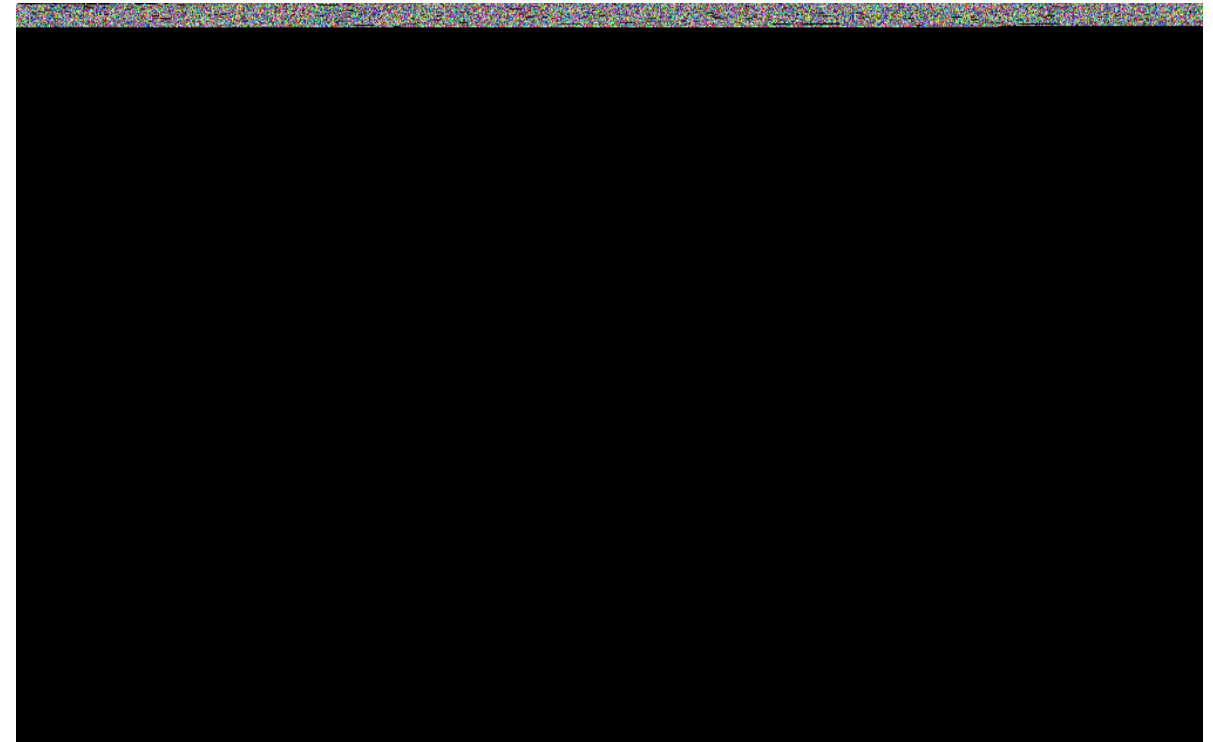
TRC elements and structures: Introduction

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Properties of fibres vs. steel reinforcement sections drive design concepts for TRC elements and structures

Reinforcement material	Tensile strength [MPa]	E-Modulus [GPa]	Strength-to-stiffness ratio [-]	Ratio relative to steel [-]
Steel	500	210	2.4	1
Carbon	3000-4000	180-240	13-22	6-9
Glass	1000-2000	70-80	12-28	5-11
Basalt	3000-4500	90-110	27-50	11-20

Non-corrosive reinforcement enables thin-walled TRC structural geometries

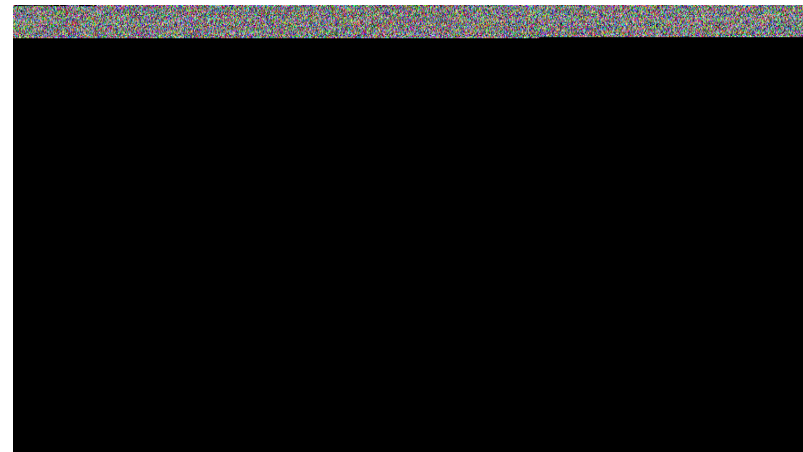
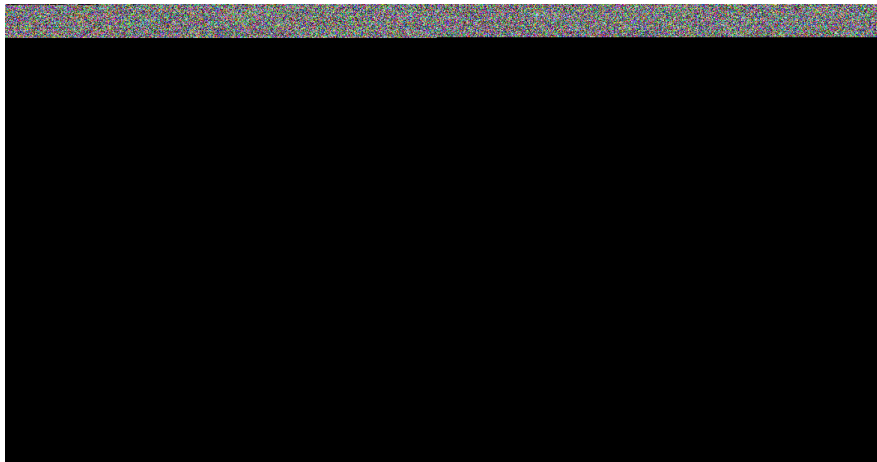


Advances in beams, slabs and bridges

TRC as a material drives towards thin section Elements
Thin-walled beams: I-beams, T-beams, U-beams
Large effective depth and hence structural stiffness
Addition of (prestressed) FRP bars in bottom
Development of design models
Applications in beams but also in bridges



Kromoser et al. (2019)



Rempel et al. (2017)

Advances in sandwich elements

Combining structural efficiency and thermal insulation:
Lightweight insulating core increases stiffness-to-weight ratio

Research to connectors between the faces

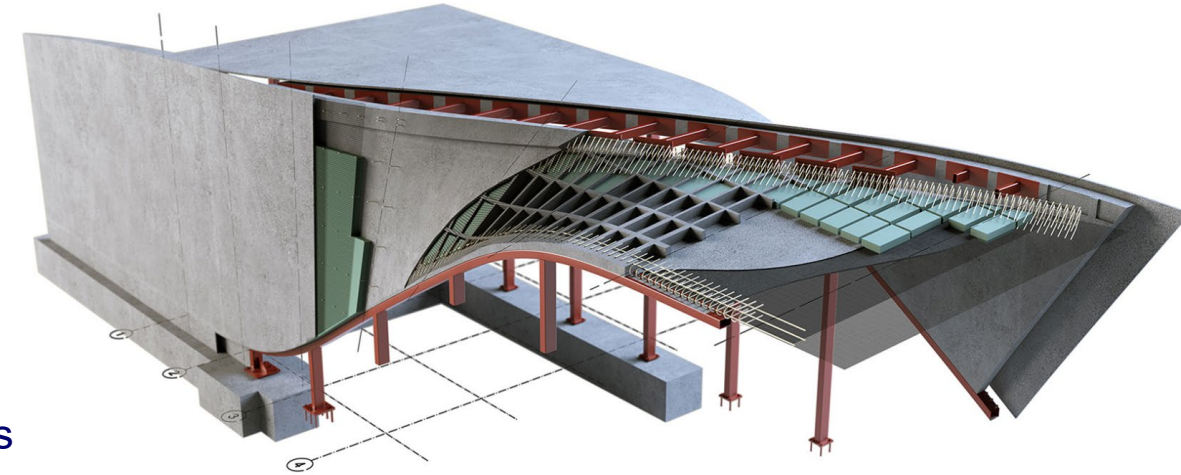
Advances in core materials:
foamed concrete, aerated concrete

From cladding/self-supporting panels
to loadbearing insulating wall elements

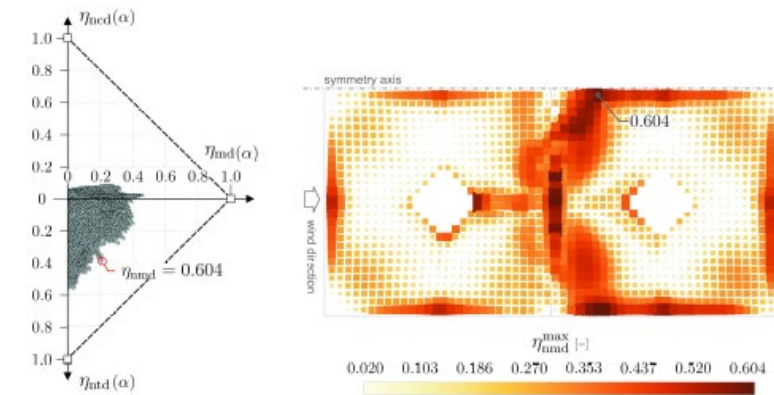


Advances in spatial shell elements

- Curved shells benefit from flexibility of textiles
- Finite element methods for modelling of nonlinear behaviour and stress redistribution
- Advances for tackling constructional aspects:
- folded shell designs, fabric formwork, modular shells,
- The CUBE Dresden showcasing twisted TRC shells
- TRC shells for (stay-in-place formwork of) vaulted floor systems



Tietze et al. (2022)



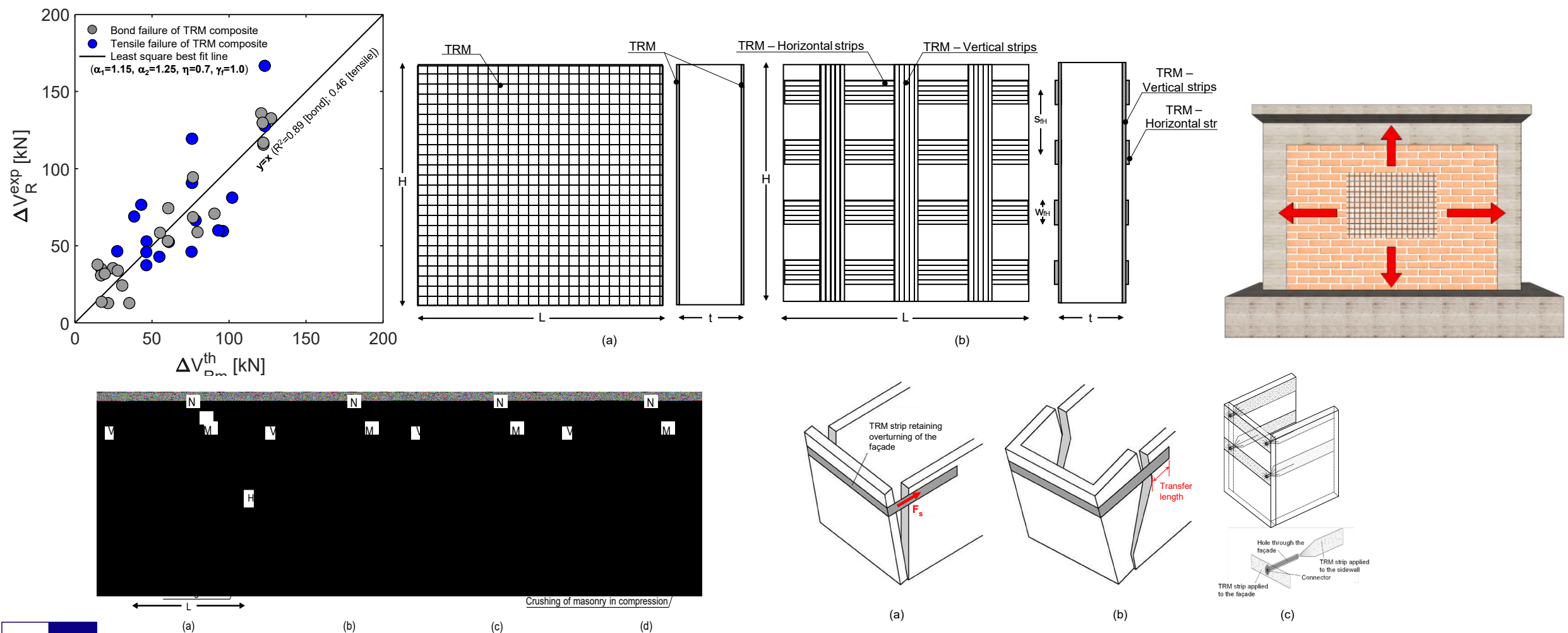
Scholzen et al. (2015)

Chapter 9: Masonry repair

- Masonry repair and strengthening: a key research area
- Externally bonded reinforcement
 - Moving from "traditional" FRPs to TRM/FRCMs: more sustainable, less hassle
 - A growing body of research has focused on TRM-strengthened masonry in the past decade
 - More focus on short-term behaviour, less on long-term performance
- RILEM & TRM/FRCMs
 - TC 201-TRC: *Textile Reinforced Concrete* – STAR published in 2006, briefly touched on the strengthening of concrete in Chapter 7
 - TC 250-CSM: *Composites for sustainable strengthening of masonry* – Concluded in 2019, first TC on TRMs/FRCMs for strengthening masonry - focused on bond performance
 - TC 290-IMC: *Durability of inorganic matrix composites used for strengthening of masonry constructions* – Coming to an end now, focused on durability of TRM-strengthened masonry
 - TC 292-MCC: *Mechanical characterisation and structural design of TRCs* - coming to an end now

Chapter 9: Masonry repair

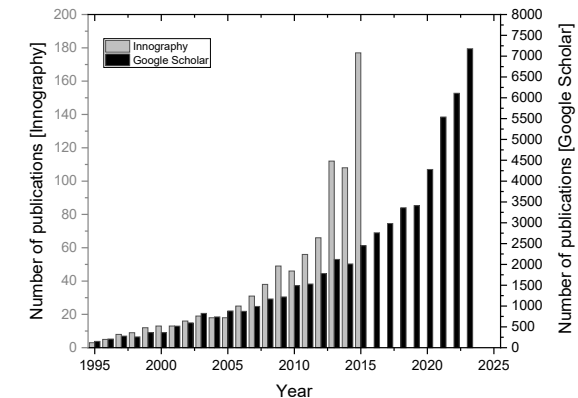
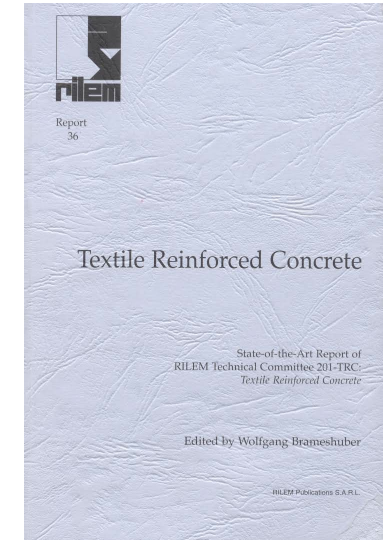
- Masonry repair chapter: ***an international collaboration of experts in the field***
 - **181 references** reviewed, **54 pages**, reflecting the state of knowledge in the field of masonry repair with TRMs/FRCMs
 - Covering:
 - Design and application concepts
 - An extensive database of experimental test results from the literature
 - In-plane and out-of-plane performance of TRM-strengthened walls
 - Performance of TRM-strengthened domes and curved elements
 - Performance of TRM-strengthened infills
 - Performance of TRM-strengthened columns
 - Combined seismic and energy retrofitting with TRMs
 - Design formulations and their development background



Key Metrics in Sustainability

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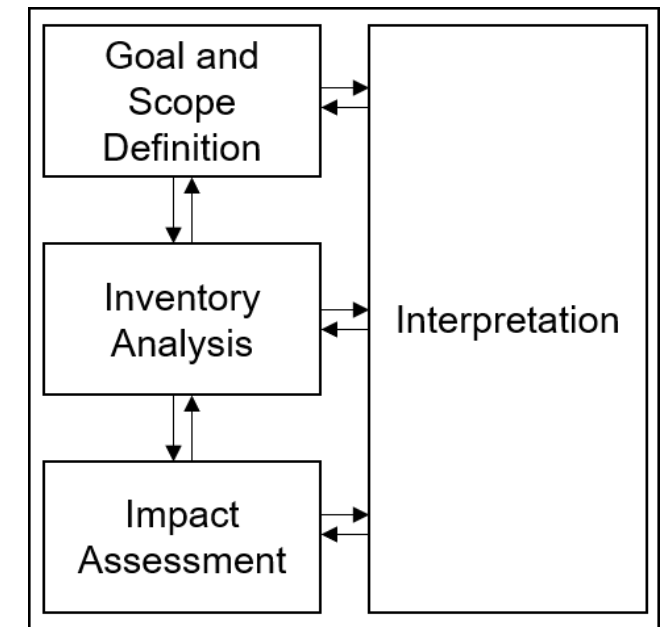
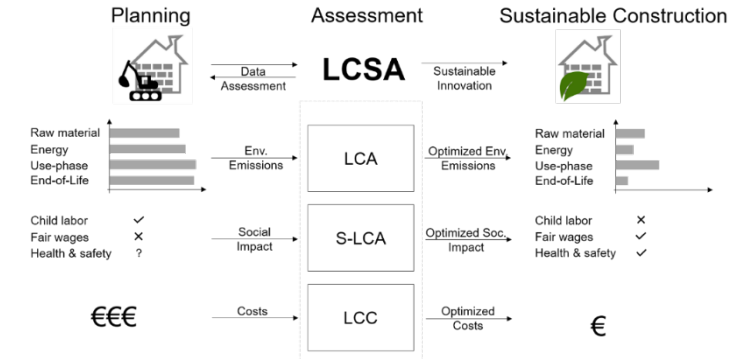
- 20 years after the first TRC State-of-the-Art report, the focus has shifted from durability and material savings towards a broader view on sustainability of Textile Reinforced Concrete
- Sustainability is now much more prominent in construction, reflected in a growing number of publications, **however**, the term is still often used too narrowly (e.g. only CO₂) or as vague marketing.
- This chapter aims to give a comprehensive state-of-the-art overview on sustainability assessment of TRC as an innovative construction material and technology.
- Structure of the chapter:
 - Basics of sustainability principles
 - Treatment of sustainability and TRC in codes and regulations
 - Guidance for Life Cycle (Sustainability) Assessment, incl. an environmental database for TRC materials and comparative LCA studies vs. steel-reinforcement
 - End of life phase of TRC – a holistic view beyond the stages “Gradle-to-gate”
- Key message of the LCAs: the result strongly depends on goal and scope definition and on the chosen impact categories – there is no absolute “good/bad” or universally “better than”.



Three dimensions of sustainability

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- Life Cycle Sustainability Assessment (LCSA) combines three equal sustainability dimensions – environmental (LCA), economic (LCC) and social (S-LCA) – based on common system boundaries and a shared functional unit.
- LCSA is increasingly used in construction because the sector has very high resource, energy and CO₂ impacts; robust life-cycle based decisions are essential, especially under frameworks like the EU Green Deal and building rating schemes (DGNB, LEED, BREEAM).
- Environmental Life Cycle Assessment (LCA) is the most mature pillar, standardized in ISO 14040/44, with four phases (goal & scope, inventory, impact assessment, interpretation) and multi-criteria indicators (e.g. GWP, resource use, toxicity).
- Life Cycle Costing (LCC) evaluates economic performance over the entire life cycle (from initial investment to maintenance and end-of-life), with different archetypes and is recommended for buildings in current standards and certification systems.
- Social Life Cycle Assessment (S-LCA) is newer and methodologically less developed; it assesses social impacts along the life cycle for defined stakeholder groups



- Sustainability in construction is increasingly driven by frameworks of codes, standards and certification systems that introduce life-cycle based, performance-oriented criteria rather than purely prescriptive rules (e.g. w/c, minimum cement content).
- For reinforced concrete, the huge global concrete volume makes optimization of embodied energy and CO₂ (through material choice, design optimization and durability) as important as reducing operational energy; TRC contributes by enabling much thinner, lighter, non-corrosive structures with significantly reduced concrete volume.
- At regulatory level, sustainability is embedded via EU CPR basic requirements, Eurocodes/EN 206, fib Model Code 2020 (which treats sustainability as an overarching design principle), CEN/TC 350 standards (harmonized methods for environmental, social and economic assessment), and ISO 14040-series plus related ISO standards and building rating schemes (DGNB, LEED, BREEAM).
- For TRC specifically, key sustainability principles derived from these frameworks are: resource conservation and recycling, structural optimization (using TRC's slenderness potential), durability and life-time engineering, and systematic life-cycle assessment, ideally supported by performance-based rather than purely prescriptive standards.

Methodological basis:

- Cradle-to-gate assessment (A1–A3) per ÖNORM EN 15804; indicators: GWP, AP, ADPf; data from EPDs & LCI (quality must be critically reviewed).

Material level findings:

- Carbon fiber production dominates CFRP impact; per kg, steel reinforcement shows lower environmental impact than CFRP.

Performance-adjusted comparison:

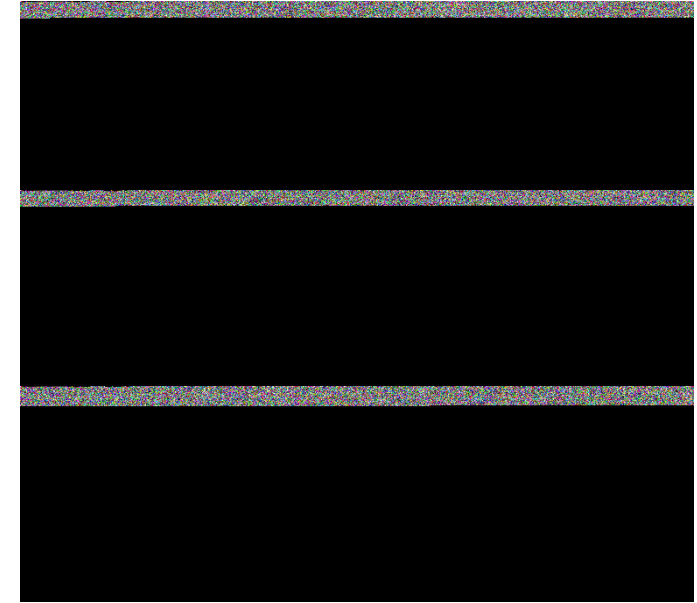
- When normalized by tensile strength and density, CFRP shows lower GWP and ADPf, reducing steel's apparent advantage.

Structural bridge case study (15 m span):

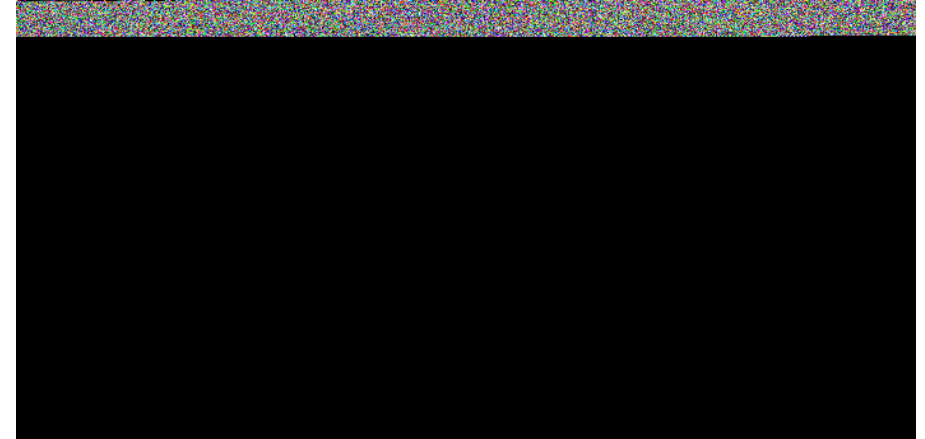
- Carbon concrete bridge shows lower GWP & ADPf than reinforced concrete and mild steel bridges; AP higher due to CFRP production; concrete mass remains main GWP driver.

Rehabilitation case – Water Tower (Malta):

- UHPC + TRC strengthening enabled conservation of heritage structure; LCA shows material production phase dominates impact, especially TRC tank strengthening—yet extends service life and avoids demolition.

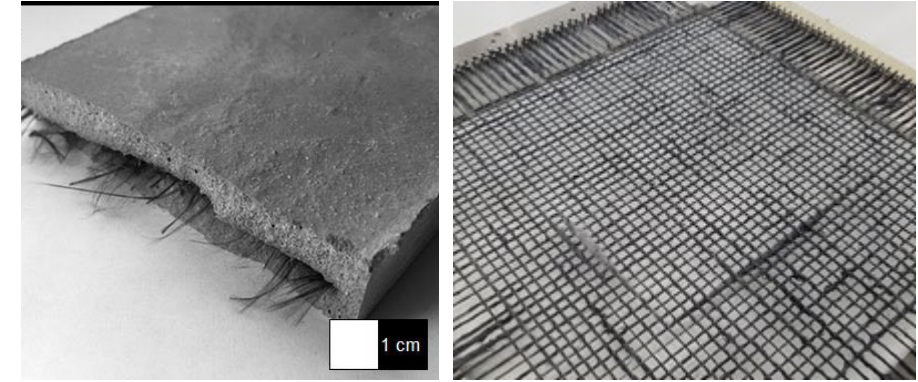


- Critical but underrepresented in LCAs – recyclability determines true sustainability
- Key sustainability dimensions:
 - Environmental: separation of carbon fibers from concrete essential
 - Economic: recycling viability depends on technology & labor costs
 - Sociocultural: health & safety concerns influence acceptance
- Separation behavior depends on:
 - Fiber type & impregnation (epoxy best recoverability; inorganic worst)
 - Textile structure (larger yarn cross-sections more robust)
 - Reinforcement position (near-surface easier to recover)
- Separation technologies:
 - Hammer mill → highest fiber exposure (~90%) but high textile damage
 - Electro-dynamic crushing → promising, lower energy demand
 - Camera-based sorting → up to ~98% fiber separation efficiency



End of life phase of TRC

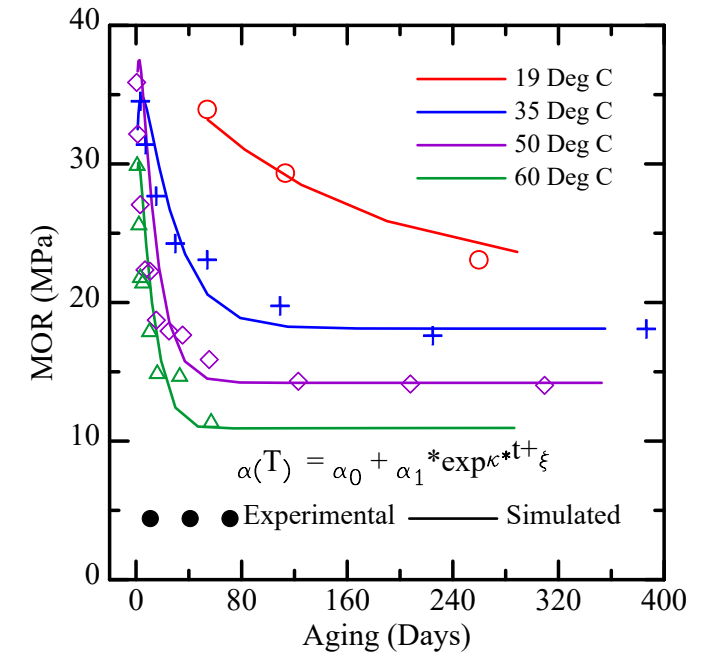
- Direct reuse of textiles not feasible (25–40% strength loss → downcycling unavoidable)
- Carbon fibers (economically viable) vs. glass fibers (not feasible)
- Recycling pathways for carbon fibers:
 - Short fiber reinforcement (0.5–1.5 vol%; +11–40% flexural strength)
 - Nonwovens (cost-effective surface reinforcement)
 - Recycled yarns (higher mechanical performance; still under research)
- Concrete fraction:
 - Can meet recycled aggregate standards (RC90)
 - Suitable for new high-performance concrete (cradle-to-cradle potential)
- Health assessment:
 - No WHO-classified hazardous carbon fibers detected
 - Risks mainly limited to skin irritation; comparable to conventional concrete processing



Durability

- One of the key aspects for the safe design of TRCs
- While the mechanics of TRCs have been the subject of several studies in the past decades, their durability has attracted less attention.
- RILEM & TRCs durability
 - TC 201-TRC: *Textile Reinforced Concrete* – STAR published in 2006, Chapter 6.3 on durability.
 - TC 290-IMC: *Durability of Inorganic Matrix Composites used for Strengthening of Masonry Constructions* (coming to an end now) – STAR to be published on durability of TRMs for strengthening of masonry / conducted an extensive inter-laboratory testing on alkali resistance of TRMs/FRCMs
 - TC 292-MCC: *Mechanical characterisation and structural design of TRCs* (coming to an end now) – STAR to be published containing a durability chapter

- Durability concepts and mechanisms
- TRC matrix durability
- Bond durability
- Protection of steel reinforcement with TRM repairs
- Embrittlement mechanisms
- An extensive database of experimental data from the literature
- Self-healing TRCs
- Developments in durability modelling
- State of codes and design standards



Chapter Leads, Team Members, (2019 – 2026)

WG 1 - Materials and Materials System (Alva Peled, Flávio Silva)

WG 2 - Constitutive Modeling (Marco di Prisco, Matteo Colombo)

WG 3 - New Elements (Rostislav Chudoba, Tine Tysmans)

WG 4 – Repair and Retrofitting (Bahman Ghiassi)

WG 5 - Durability and Sustainability (Martin Hunger)

Members: 82 Members (23 nationalities) .

Compilation of Results

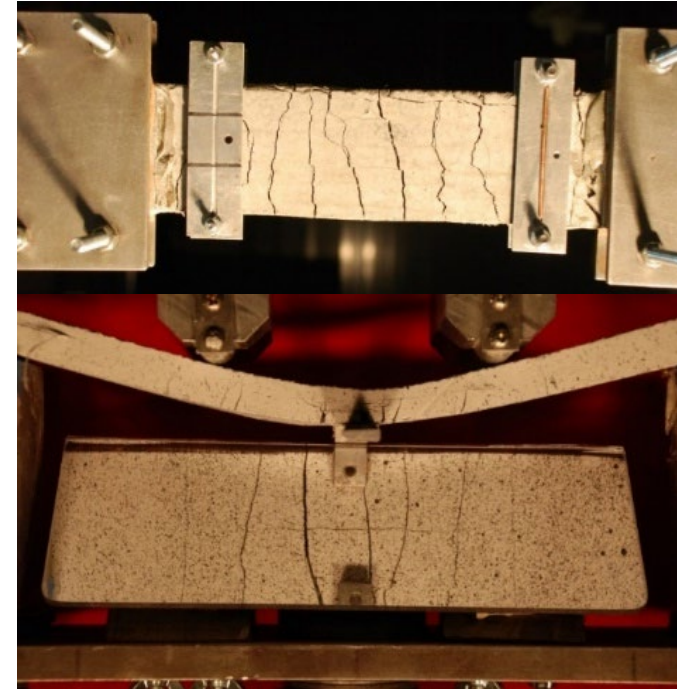
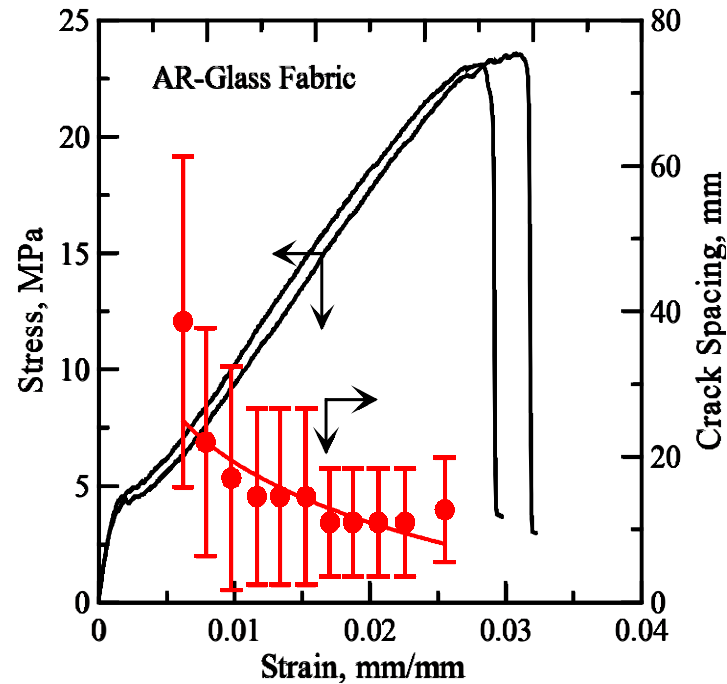
- The state-of-the-art-report – Document already submitted to Springer.
- Literature review papers:
 - Bond behaviour of textile-reinforced concrete – a review – Materials and Structures v.57, Issue 4, 2024
 - Life Cycle Assessment (LCA) of Non-metallic Reinforcement for reinforcing concrete: Manufacturing, Durability, Dismantling, Recycling and Reuse – a Review. Materials and Structures, 2023, v. 56, n.126

Presentations, Sessions, publications and networking:

- Initial meetings in Germany (Dresden) and Italy (Desenzano del Garda).
- ACI Fall Convention in Cincinnati Ohio, 2019
- ACI 2020 online Textile Reinforced Cement Composites – New Applications and Repair Materials (9 presentations) - <https://www.concrete.org/education/freewebsessions.aspx>
- ACI Special Publication on TRC
- CICE 2023 Rio de Janeiro – July 2023 –
- FRC 2023 – Arizona State University – September 2023
- BEFIB – Dresden, September 2024 – Keynote by Barzin and Tine
- Ghent – 4 Sessions on TRC, 2026

Chapter 5 – Constitutive relationships

Multiple Cracking and Crack spacing in tension and flexure



Dominant toughening mechanisms are attributed to matrix cracking, ply delamination, and crack deflection mechanisms. Effective design procedures take into account the stiffness reduction for utilization of full strain range.

Chapter 5 – Constitutive relationships

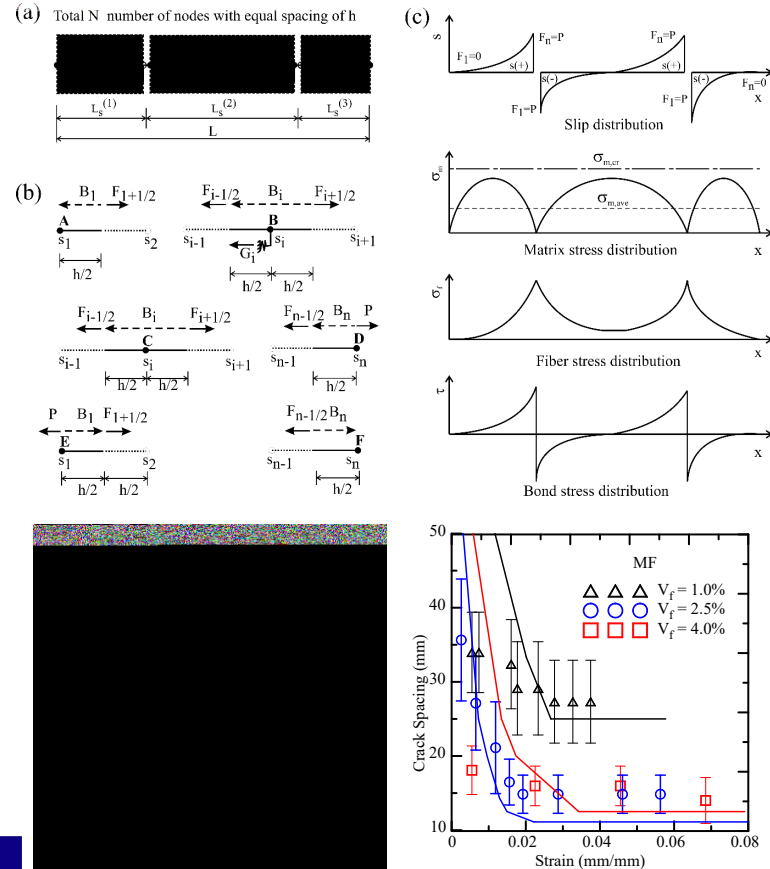
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Homogenized constitutive relationships

Lyon paper

Finite different approach: Hierarchical Modeling

Mobasher et al. 2021



The mechanical properties and damage parameters such as crack spacing and crack width can be integrated into the simulation of the composite tensile stress-strain response. A tension stiffening finite-difference method is used. The model simulates a tension specimen that is idealized as a series of 1-D segments consisting of textile, matrix, and interface elements.

The matrix is treated as brittle with no strain-softening response, and the differential slip is defined as the cumulation of strain difference between the textile and matrix.

As the applied nominal stress reaches the implied tensile strength of the matrix phase, a crack is placed at the section and the load is transferred to the textile phase through the interface elements and the bridging mechanism.

Subsequent cracks form as the debonding and pullout segments continue to transfer the load back and forth between the cracked sections and uncracked composite.